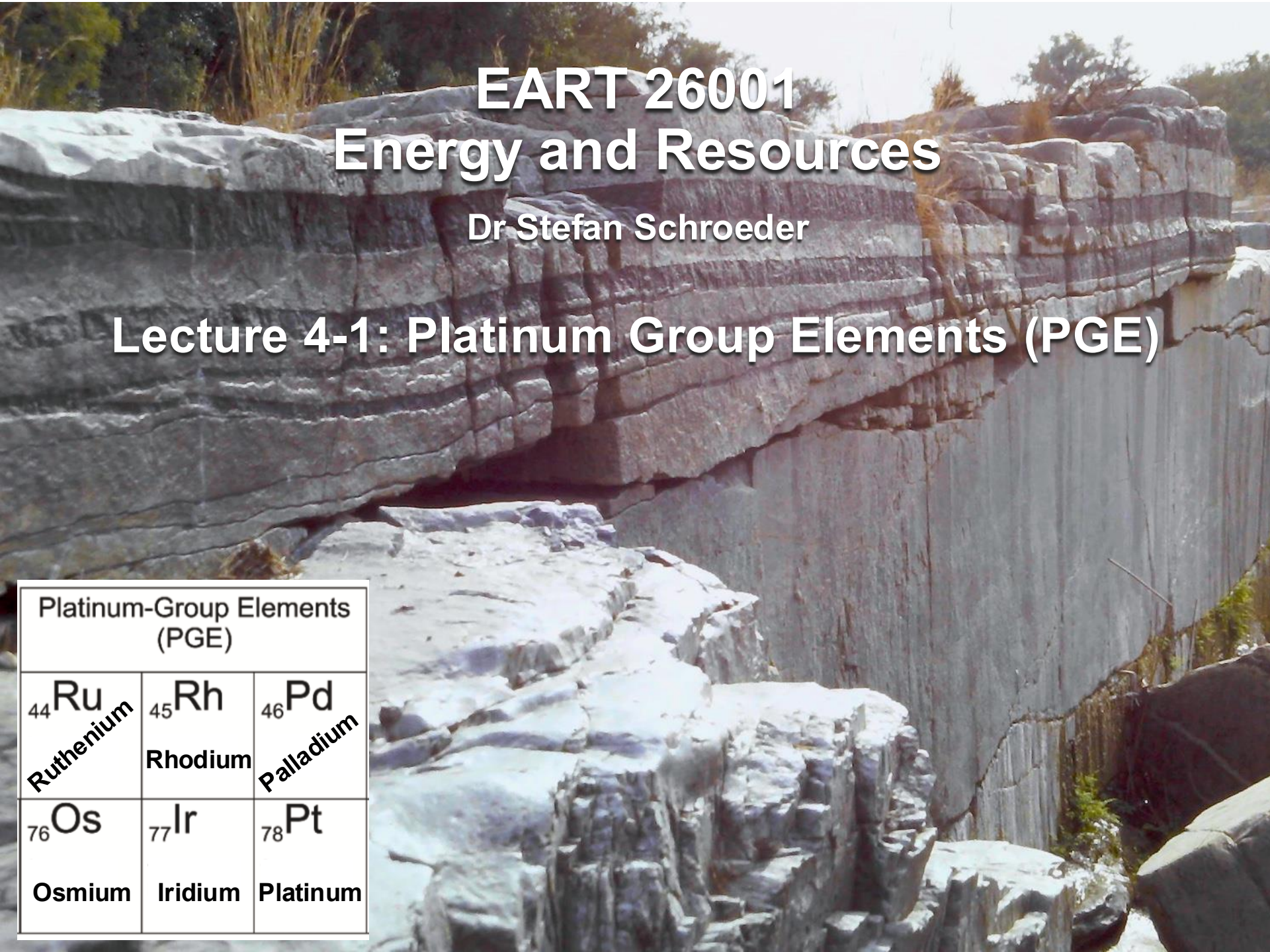
EART 26001
Energy and Resources

Dr Stefan Schroeder

Lecture 4-1: Platinum Group Elements (PGE)



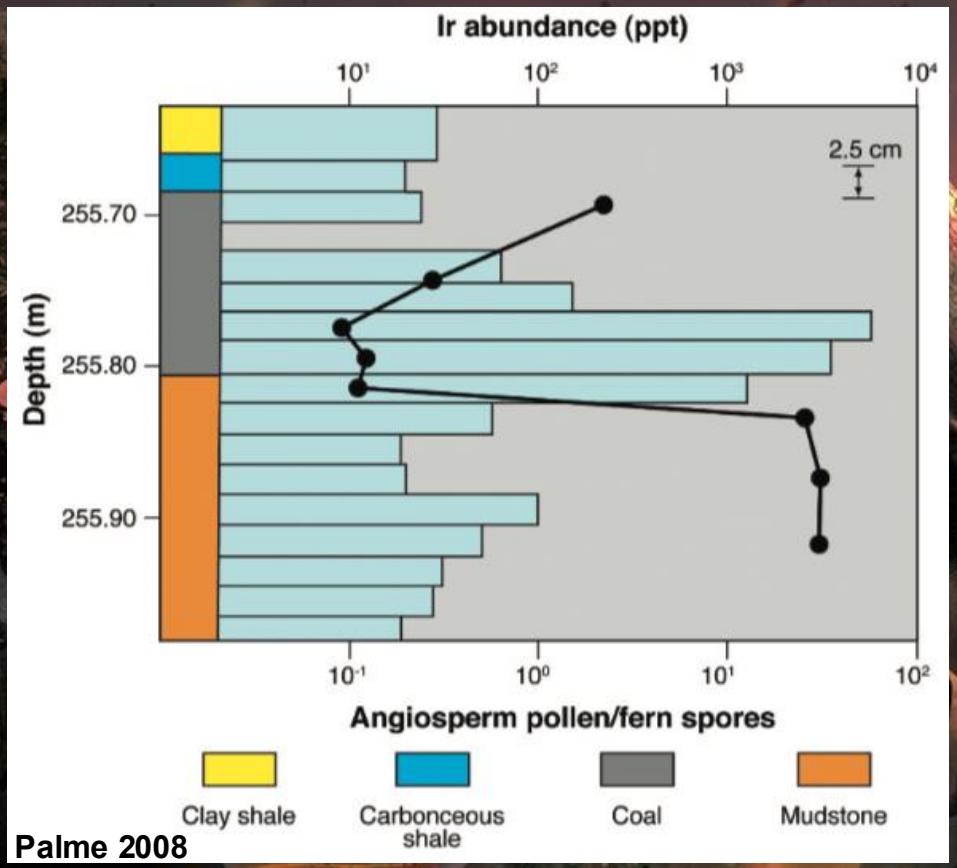
Platinum-Group Elements (PGE)		
$_{44}$Ru Ruthenium	$_{45}$Rh Rhodium	$_{46}$Pd Palladium
$_{76}$Os Osmium	$_{77}$Ir Iridium	$_{78}$Pt Platinum

Energy and Resources
Learning goals week 4

This week you will learn to:

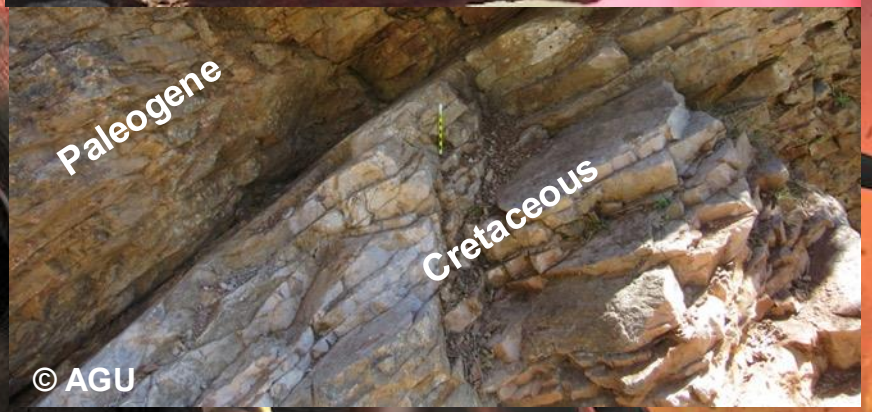
- **Describe the uses, physical and chemical properties of platinum group elements (PGE).**
- **Recognize the igneous geological processes forming PGE enrichment in magmatic resource deposits, as exemplified by the Bushveld Complex.**
- **Recognize the exploration, production methods, and environmental impact of PGEs.**

Platinum group elements
Properties and applications



Palme 2008





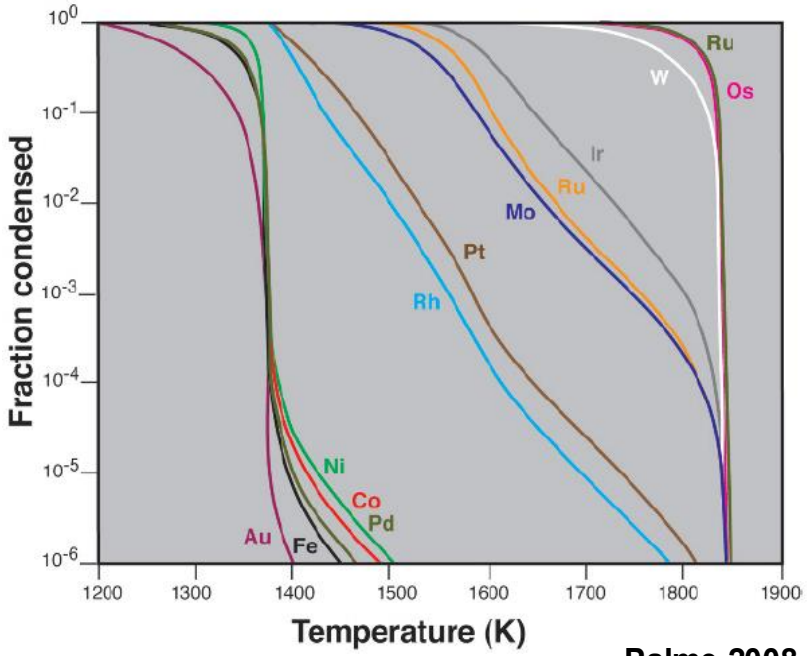
- Highly enriched in core, mantle and extraterrestrial bodies

© The Times

Platinum group elements
Properties and applications

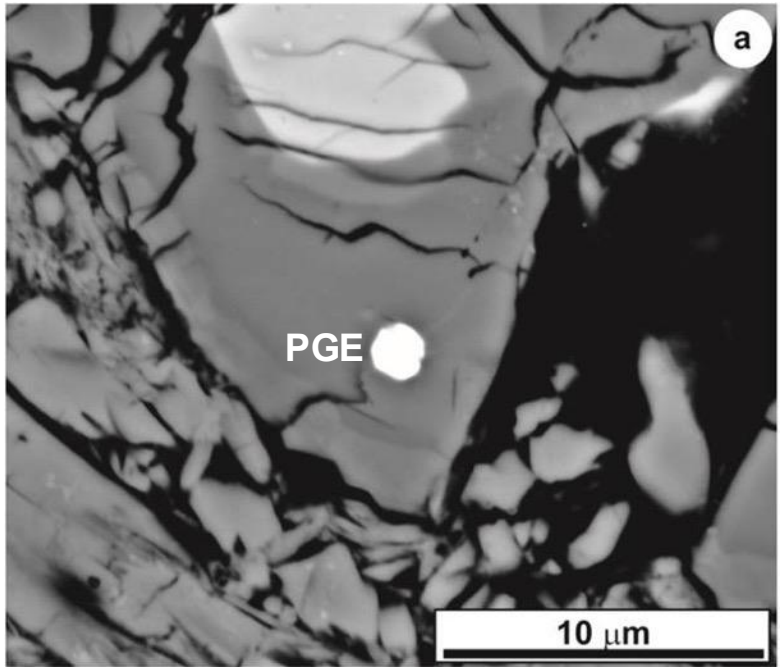
Platinum-Group Elements (PGE)				
	$_{44}$Ru hcp mp 2607 K	$_{45}$Rh fcc mp 2237 K	$_{46}$Pd fcc mp 1828 K 1555°C	
$_{75}$Re hcp mp 3459 K	$_{76}$Os hcp mp 3306 K 3033°C	$_{77}$Ir fcc mp 2739 K	$_{78}$Pt fcc mp 2041 K 1768°C	$_{79}$Au fcc mp 1337 K

- High melting and condensation temperatures
- Early condensation from solar nebula and in mantle
- Meteorite concentration 400-900 ppb
- Crustal concentration 0.4-0.5 ppb
- Siderophile: bond with iron and sulfides
- Solute transport as S and Cl complexes



Palme 2008

Allende meteorite



Campbell 2001

Platinum group elements
Properties and applications – catalytic reactions





© Royal Institution, Paul Wilkinson

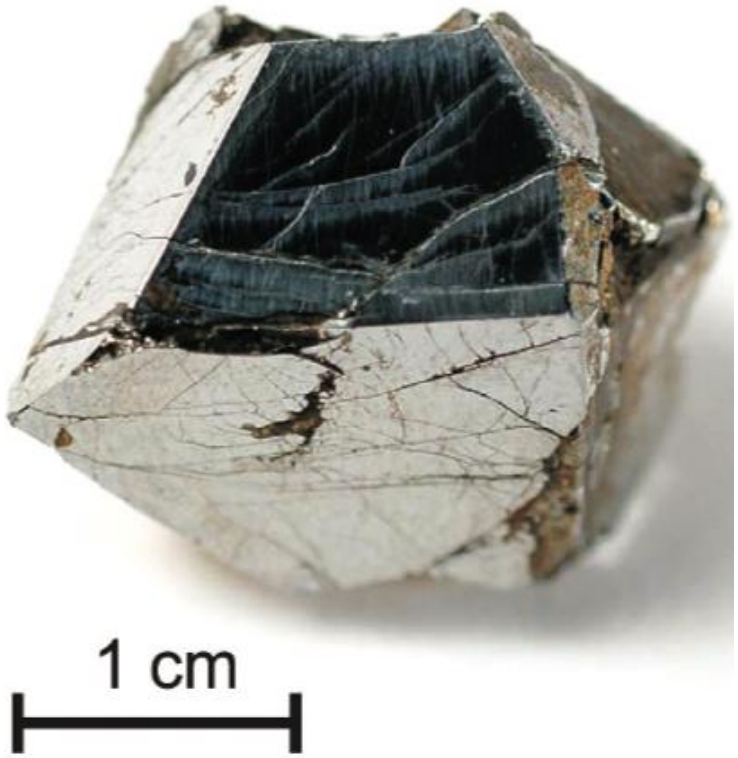


© Royal Society London

- Davy's safe lamp (1817) included a platinum wire that limited combustion of flammable coal gases: first application of a catalytic reaction

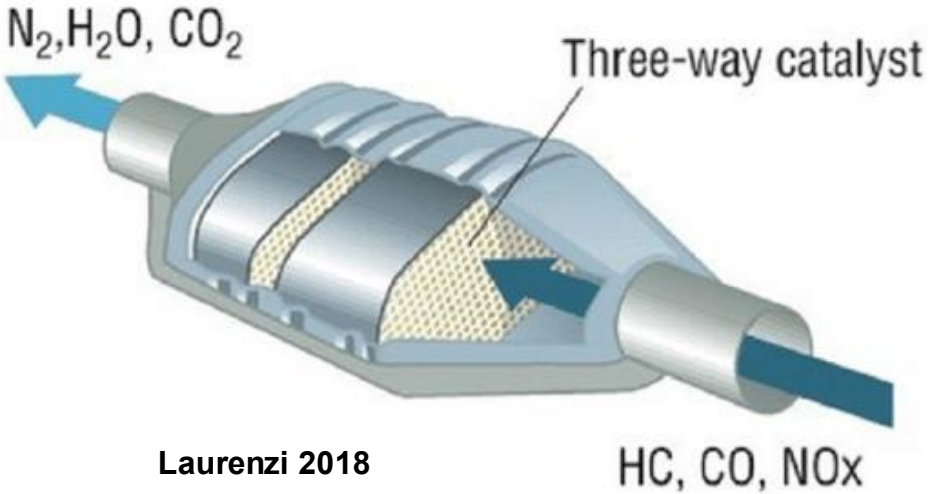
Platinum group elements
Properties and applications

- **Precious metal: resistant to corrosion and oxidation, and stable at high temperatures**



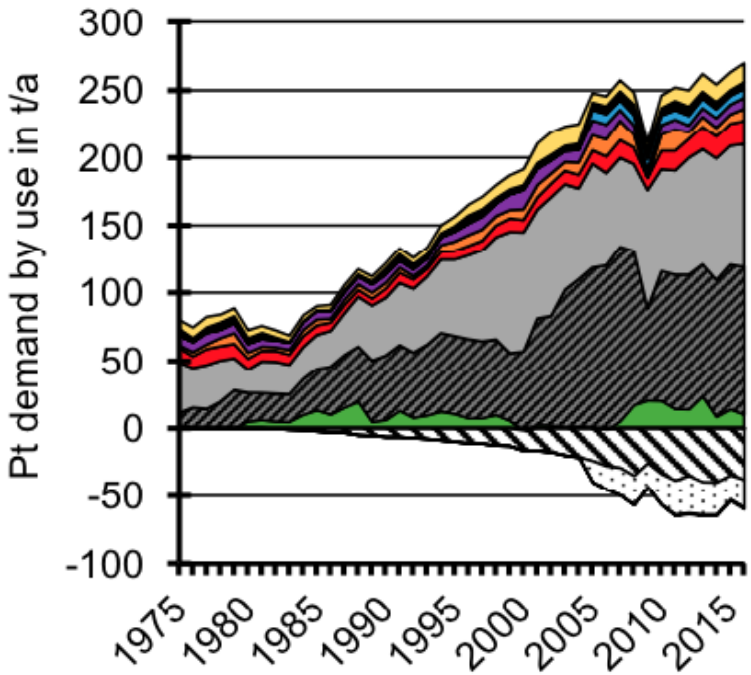
Brenan 2008

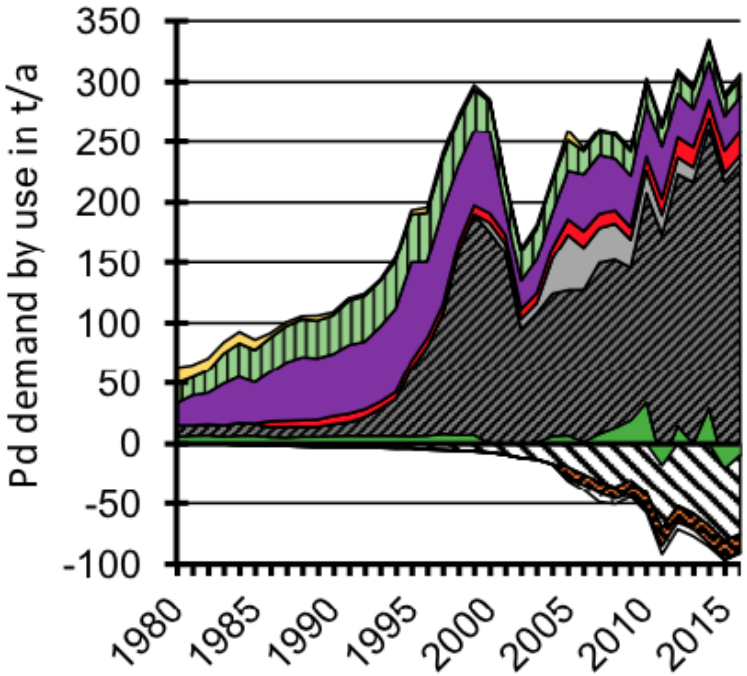
- **Catalyst in industrial applications:**
  - **Automotive: oxidation of exhaust gases**
  - **Petrochemical: improve octane rating of fuels**
  - **Chemical: production fertilizer, explosives, nitric acid, hydrogen cyanide.…**



Laurenzi 2018

Platinum group elements
Properties and applications





Thormann 2017

- Jewelry
- Electrical (contacts, wires)
- Electronics (hard drives, LCD displays
- Medical and dental

Other
Petroleum
Medical & Biomedical
Electrical
Glass
Chemical
Jewelry
Automotive exhaust gas catalytic converter
Electrochemical
Recycling: Electrical
Recycling: Jewelry
Recycling: Autocatalyst
Investment
Dental

Platinum group elements
Minerals

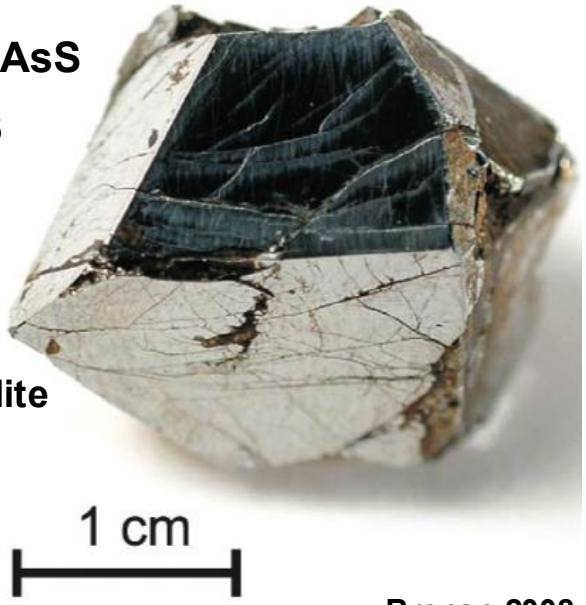
Associated with sulfides:

- **Pyrrhotite $Fe_{1-x}S$**
- **Pentlandite $(Fe,Ni)_9S_8$**
- **Chalcopyrite $CuFeS_2$**

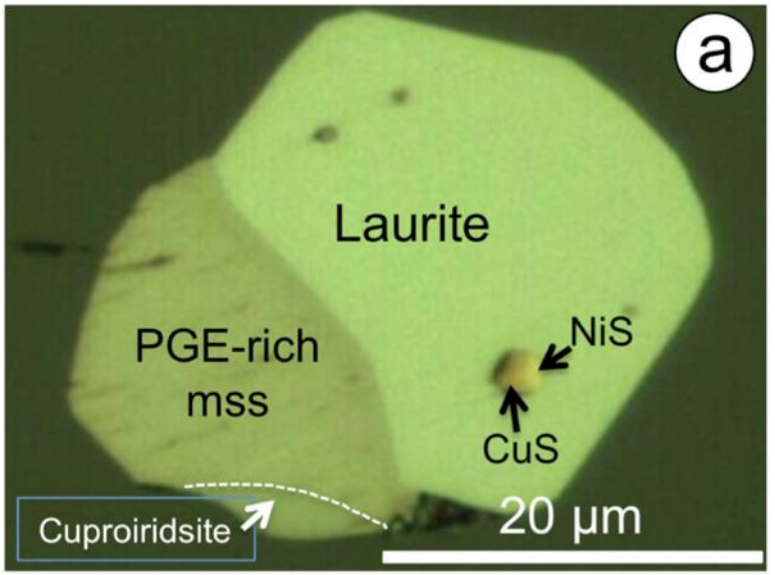
Accessory PGE minerals (PGM):

- **Sperrylite $PtAs_2$**
- **Laurite RuS_2**
- **Erlichmanite OsS_2**
- **Irarsite (Ir,Ru,Rh,Pt)AsS**
- **Braggite (Pt,Pd,Ni)S**
- **Cooperite PtS**





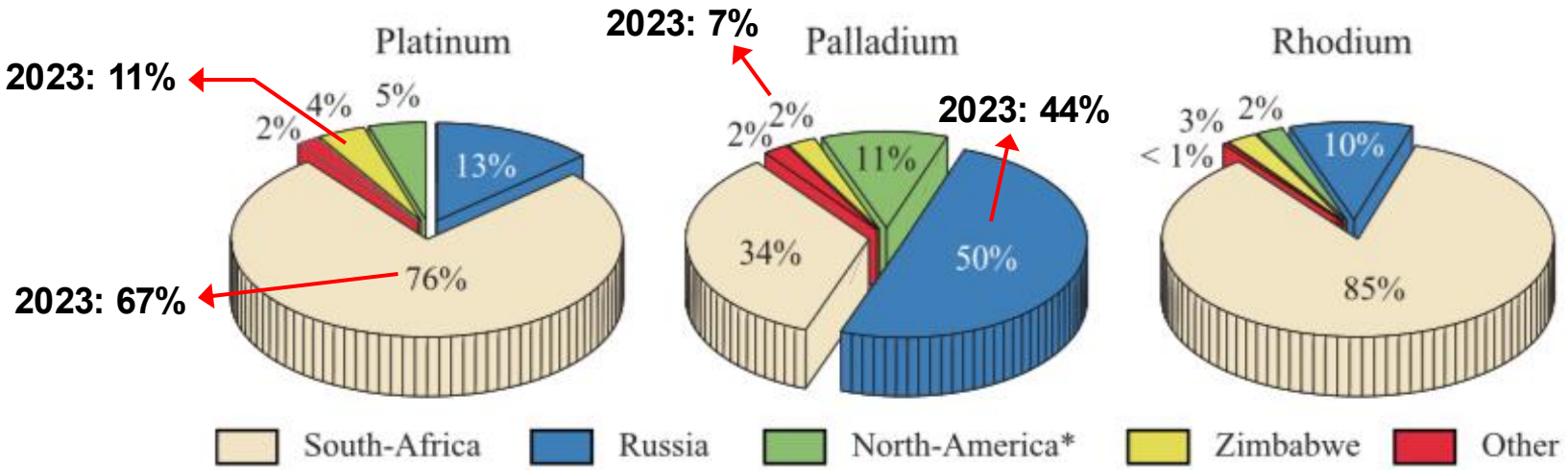
Brenan 2008



Brenan 2008

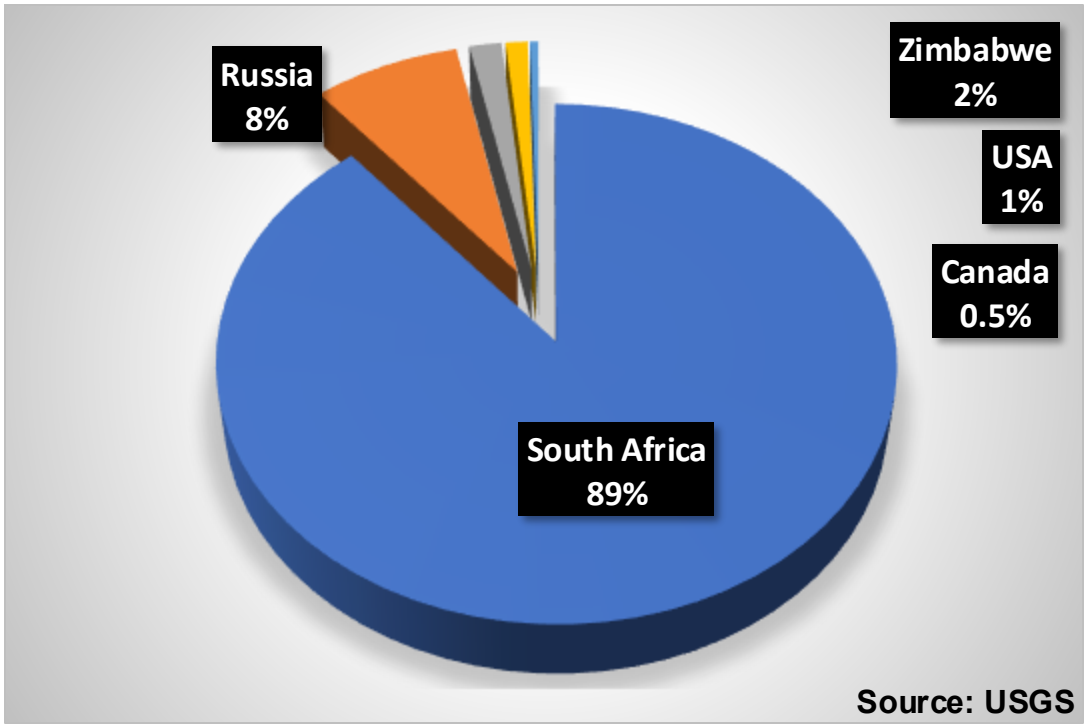
Platinum group elements
Reserves & Production

Mine production 2007-2011



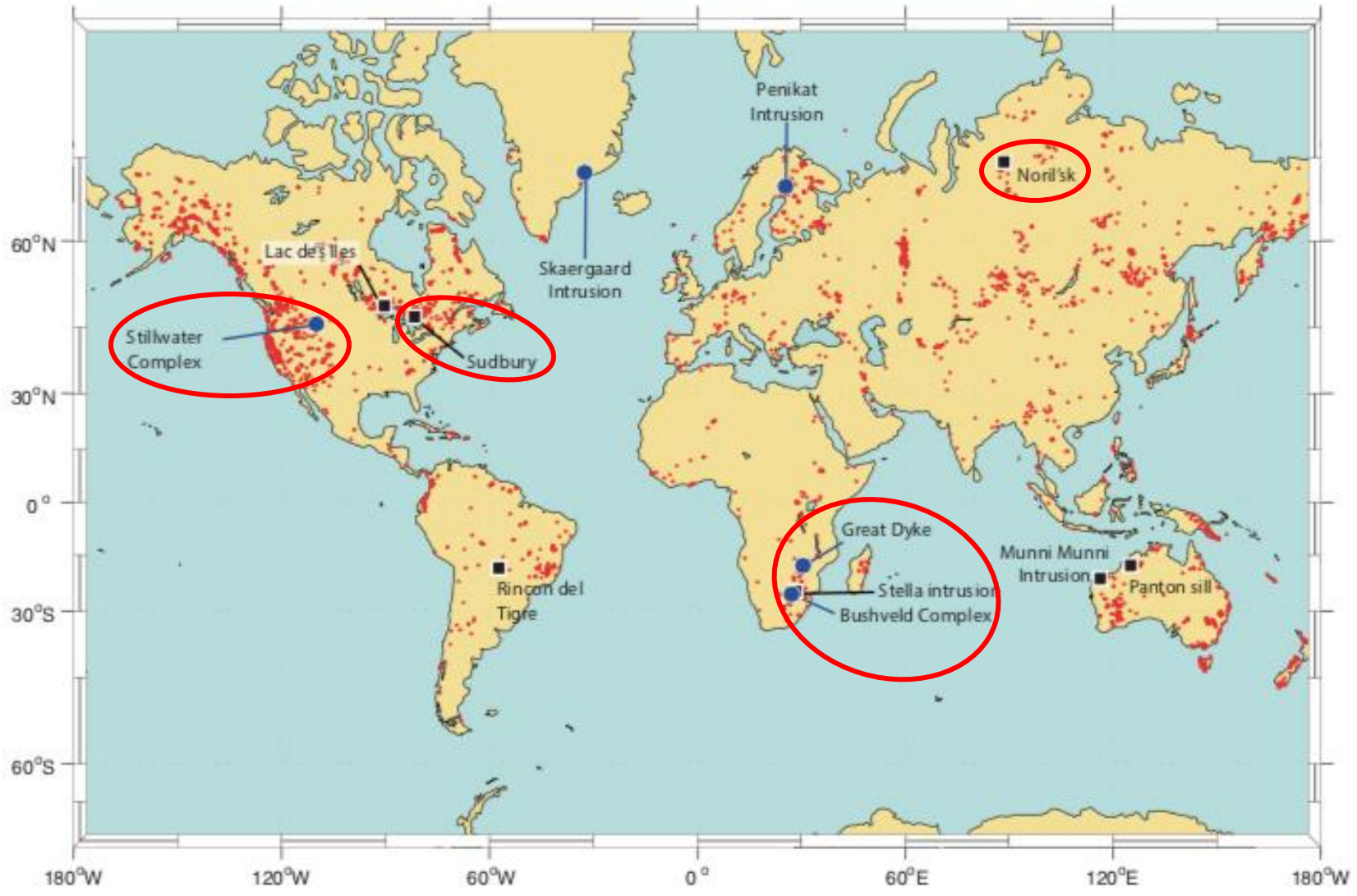
Godel 2015

PGE Reserves 2024



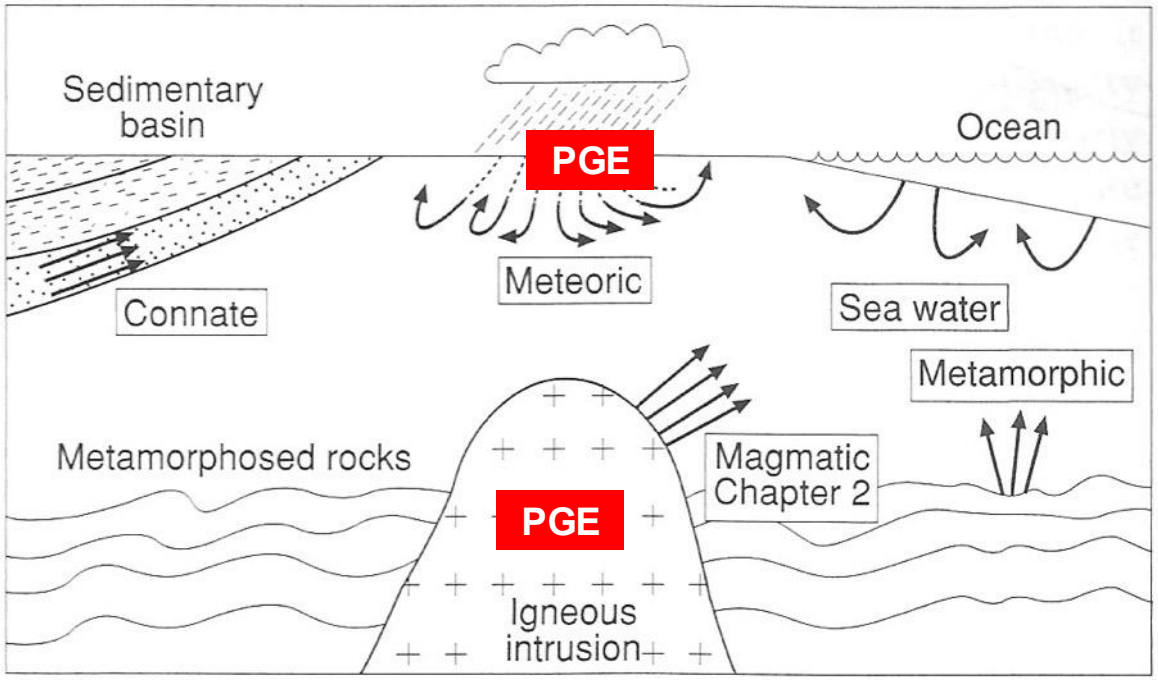
- **Economic ore grade 2.5-10 g/t (ppm), i.e. enrichment <25,000x**

Platinum group elements
Reserves & Production



Godel 2015

Platinum group elements
Deposit types



Robb 2005

- **Essentially associated with igneous intrusions (layered igneous intrusions): orthomagmatic**
- **Placers**

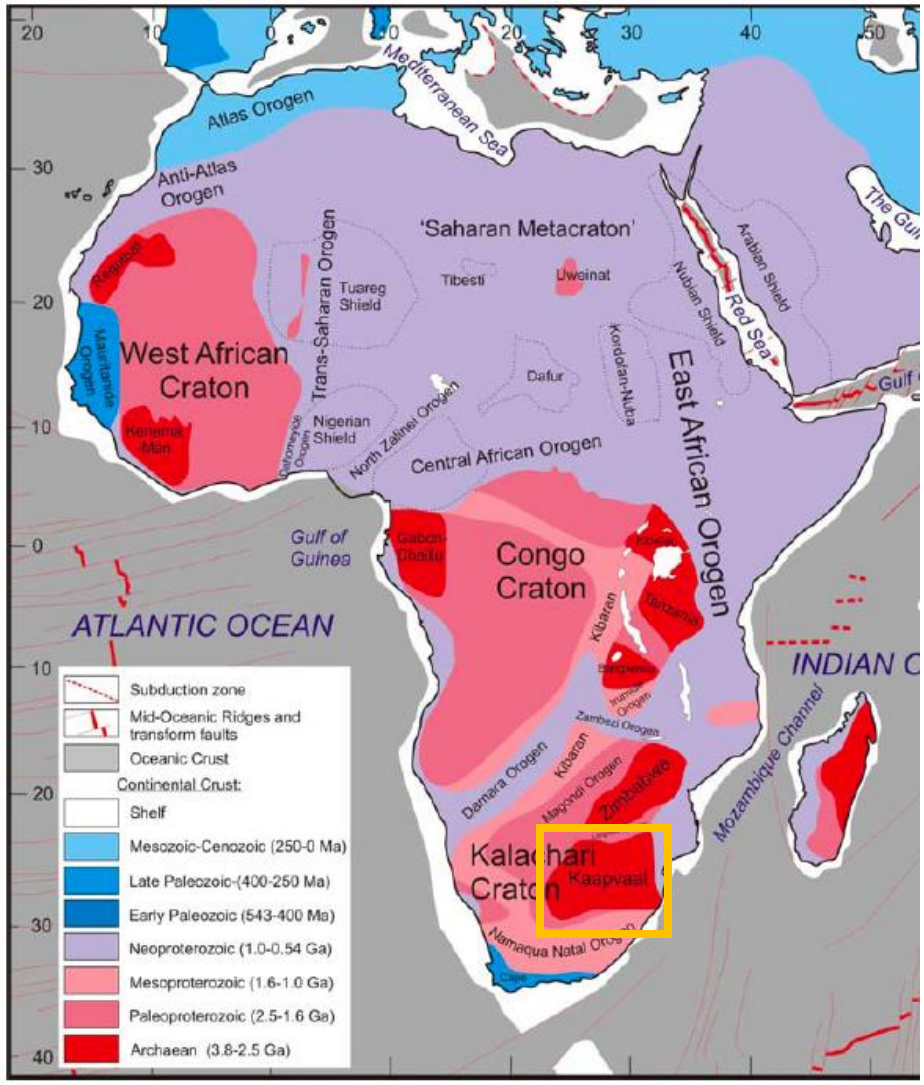


EART 26001
Energy and Resources

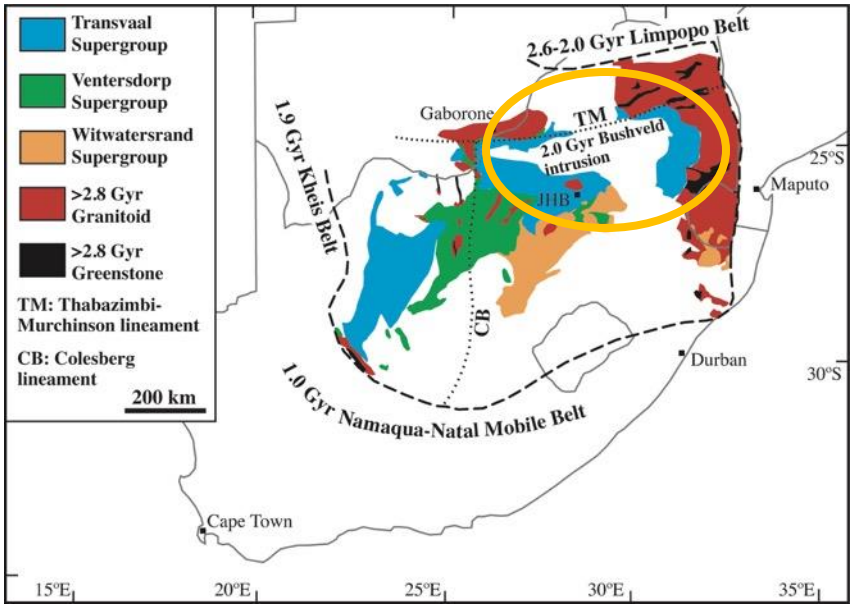
Dr Stefan Schroeder

Lecture 4-2: Layered igneous intrusions and PGE (Bushveld Complex)

Bushveld Complex
Introduction

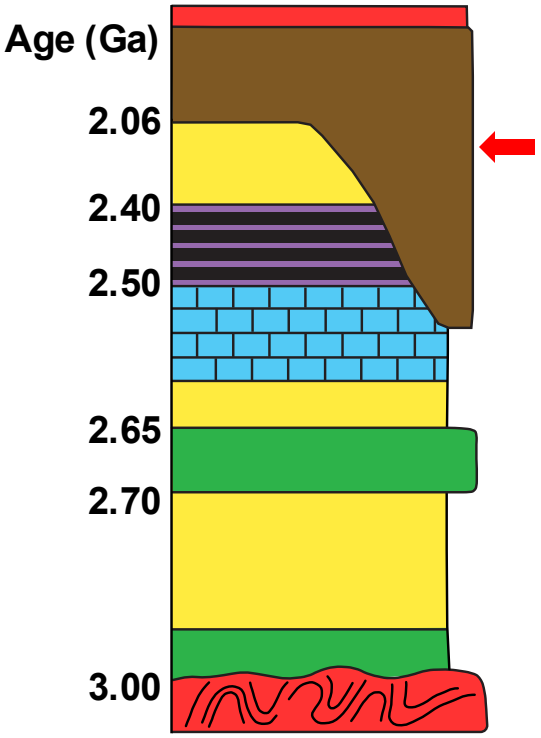


Van Hinsbergen 2011, after Gubanov & Mooney 2009

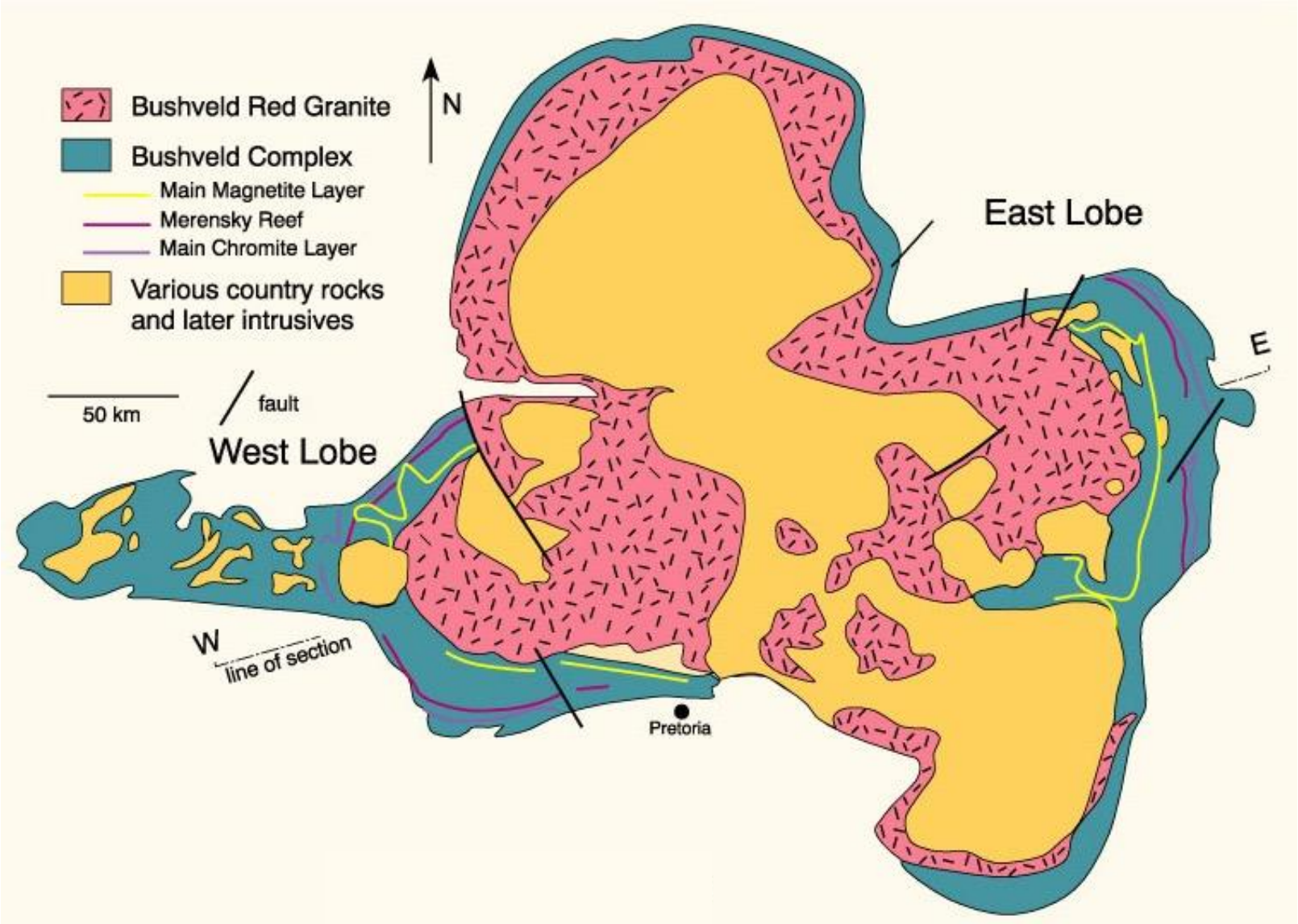


after Schmitz 2004

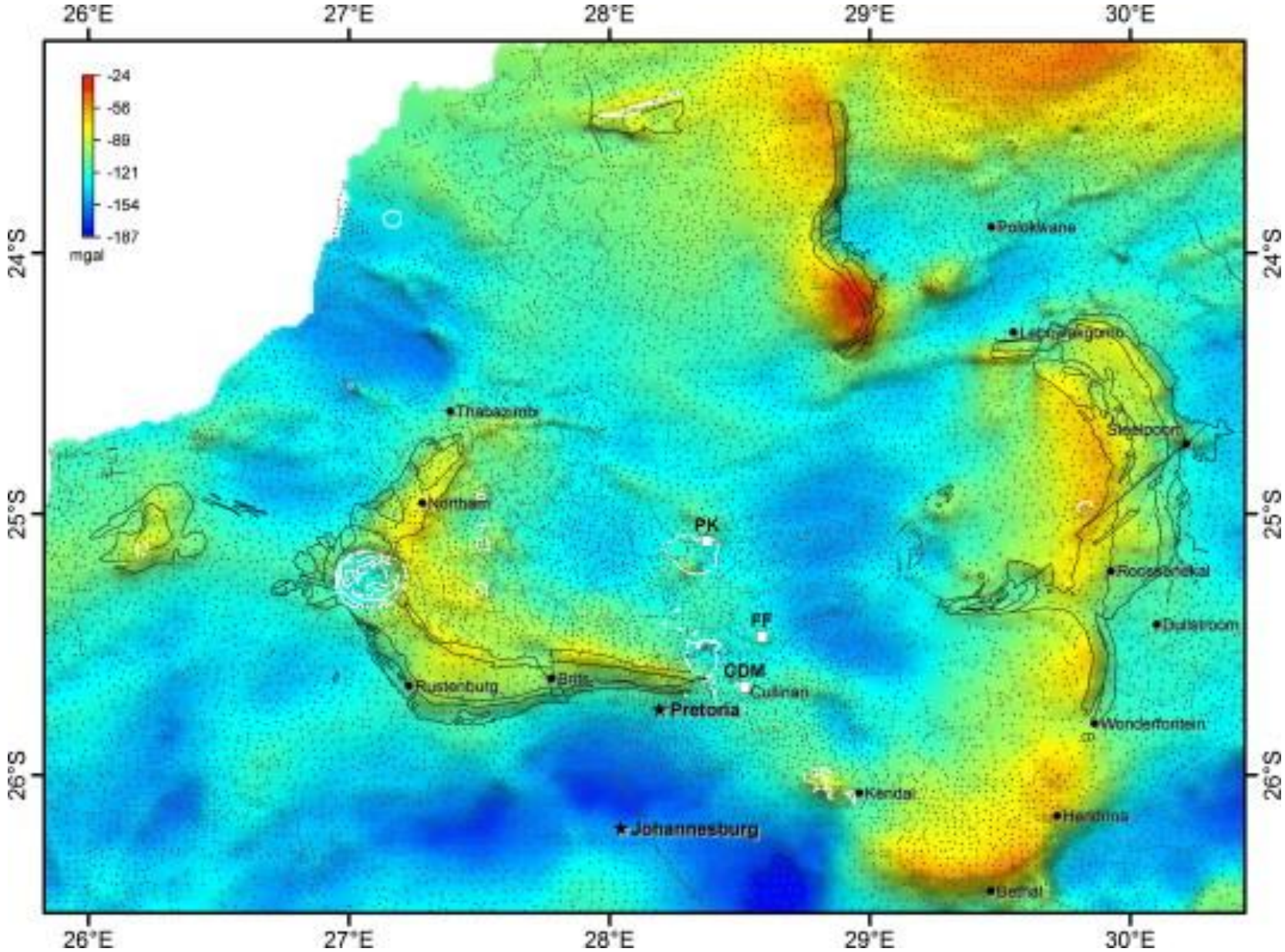
- Igneous intrusion into Archean-Paleoproterozoic sedimentary cover sequence on Kaapvaal Craton



Bushveld Complex
Structure



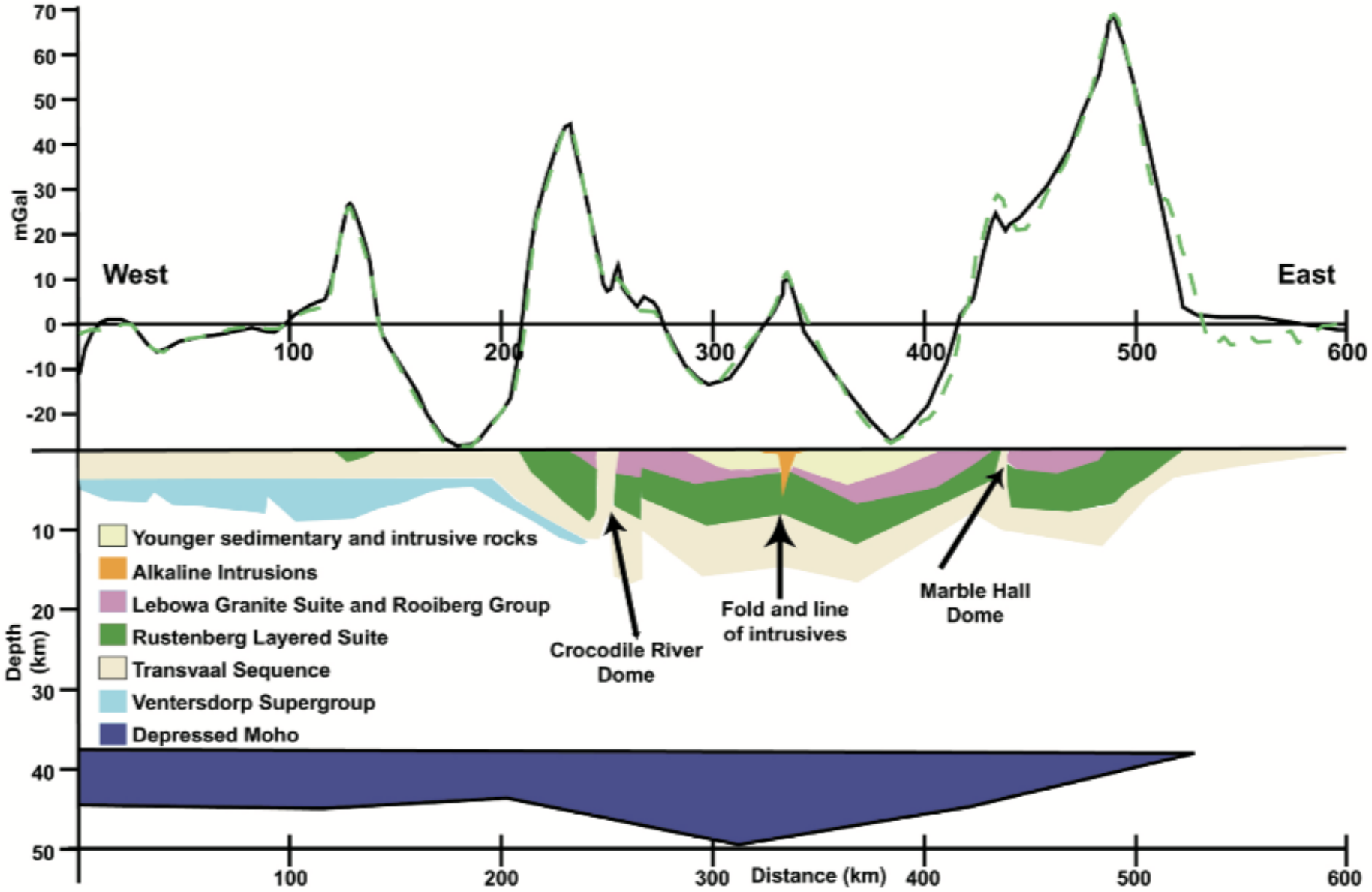
Bushveld Complex
Structure



Cole 2014

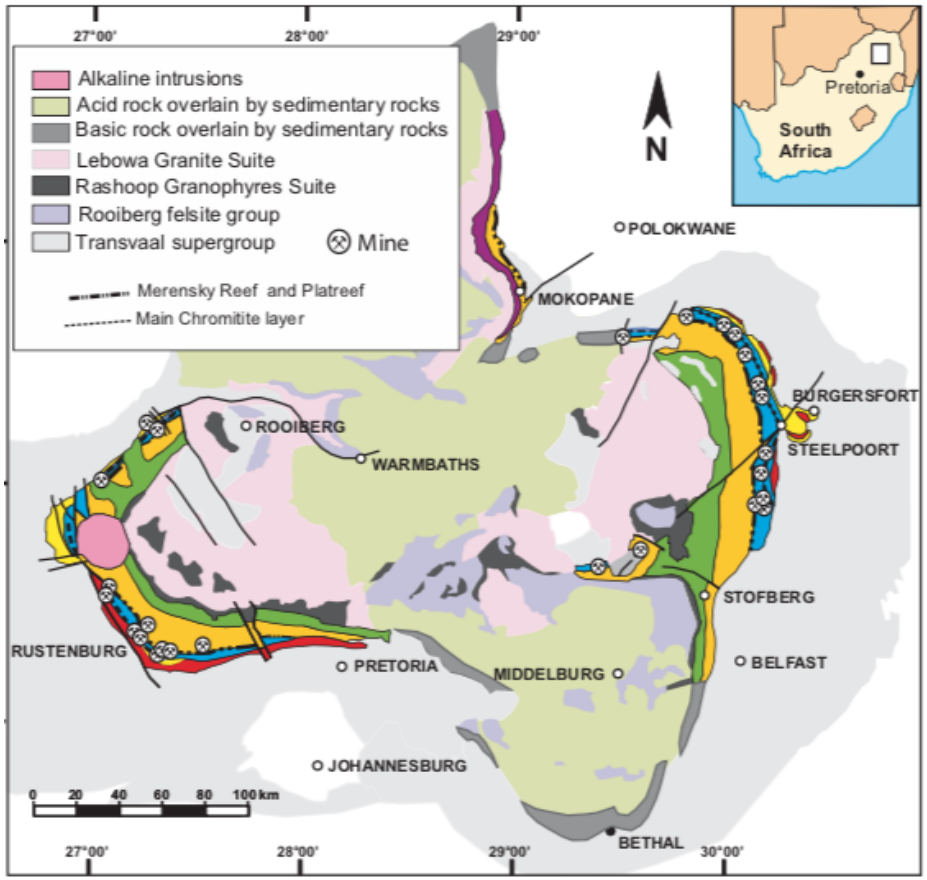
Bushveld Complex
Structure

- >70,000 km^2, several hundred km diameter
- 1 million km^3 of magma in 65,000 years



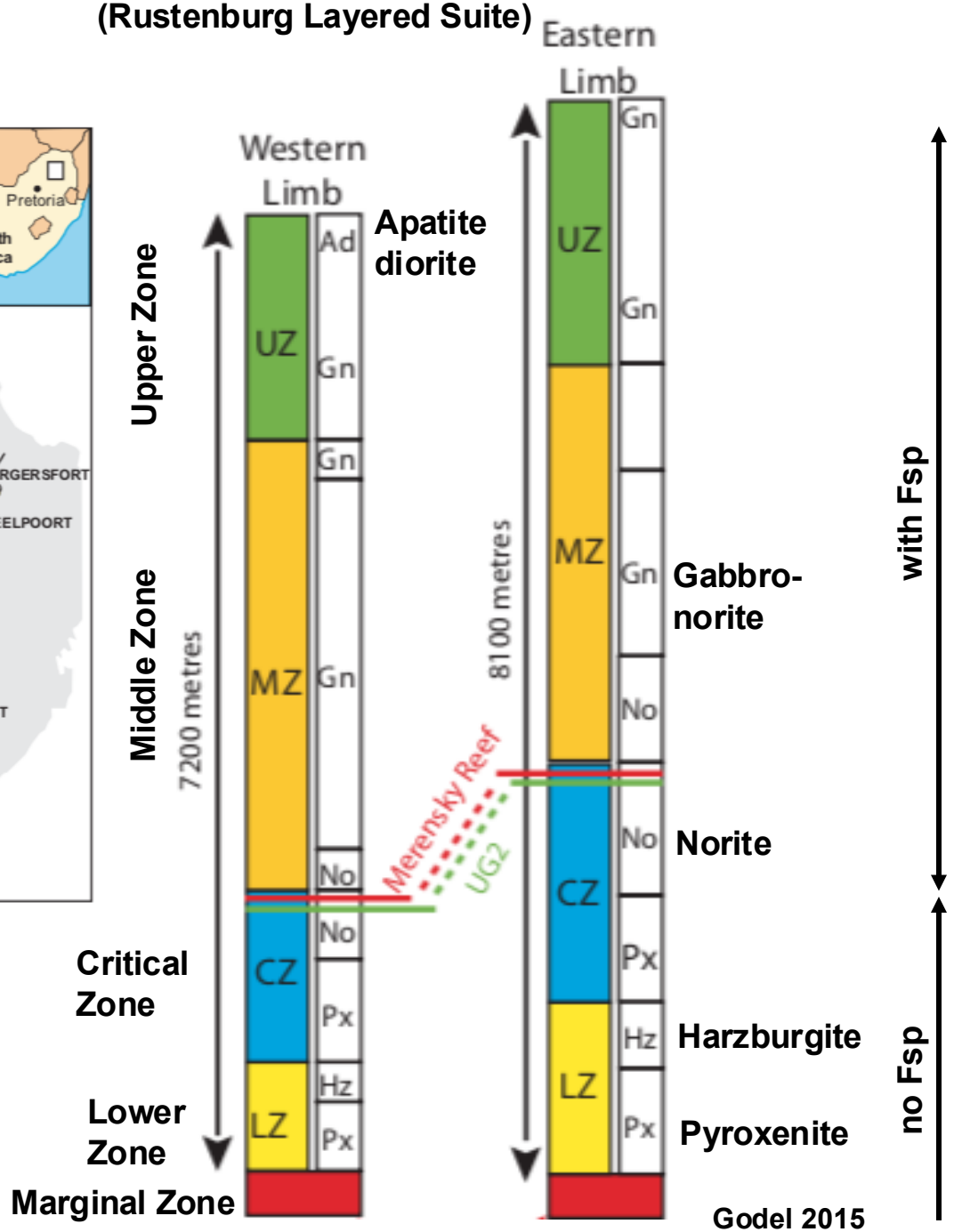
Webb 2004

Bushveld Complex
Stratigraphy

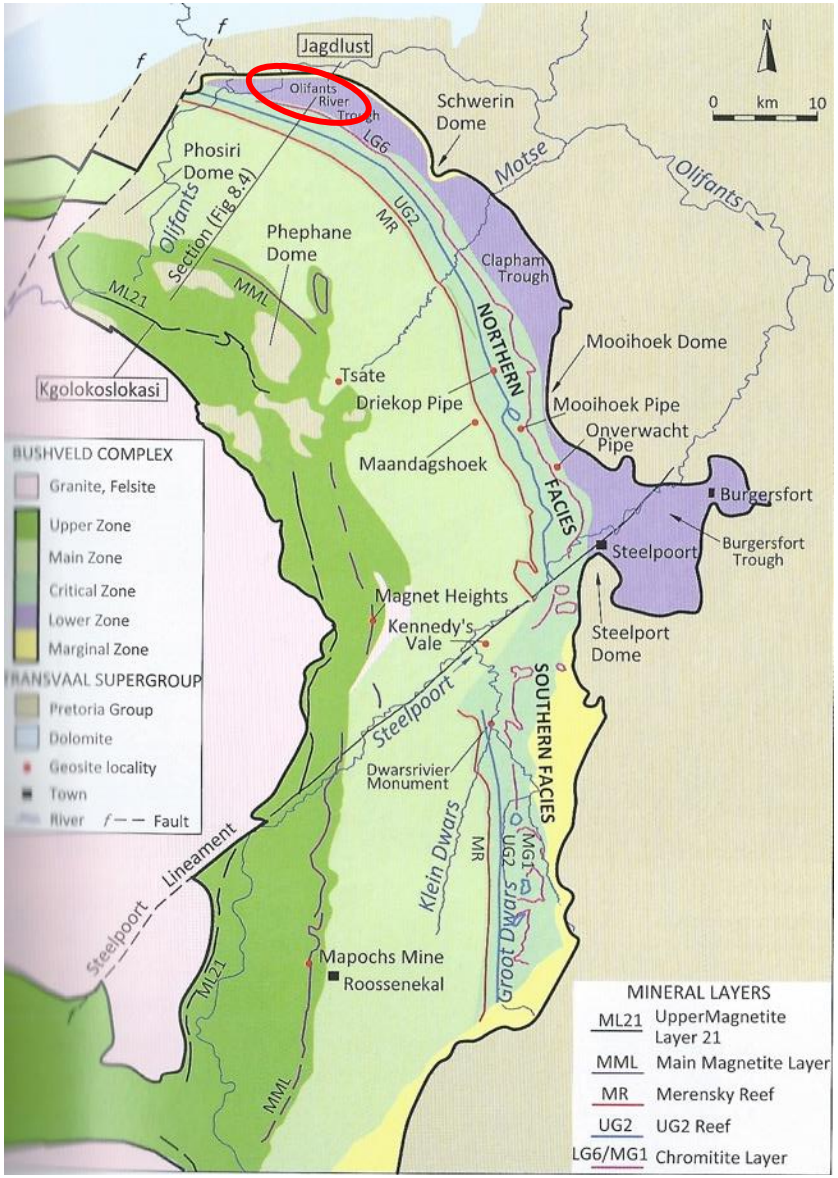


- Distinct layering of ultramafic and mafic rocks

Bushveld mafic and ultramafic rocks (Rustenburg Layered Suite)



Bushveld Complex
Stratigraphy

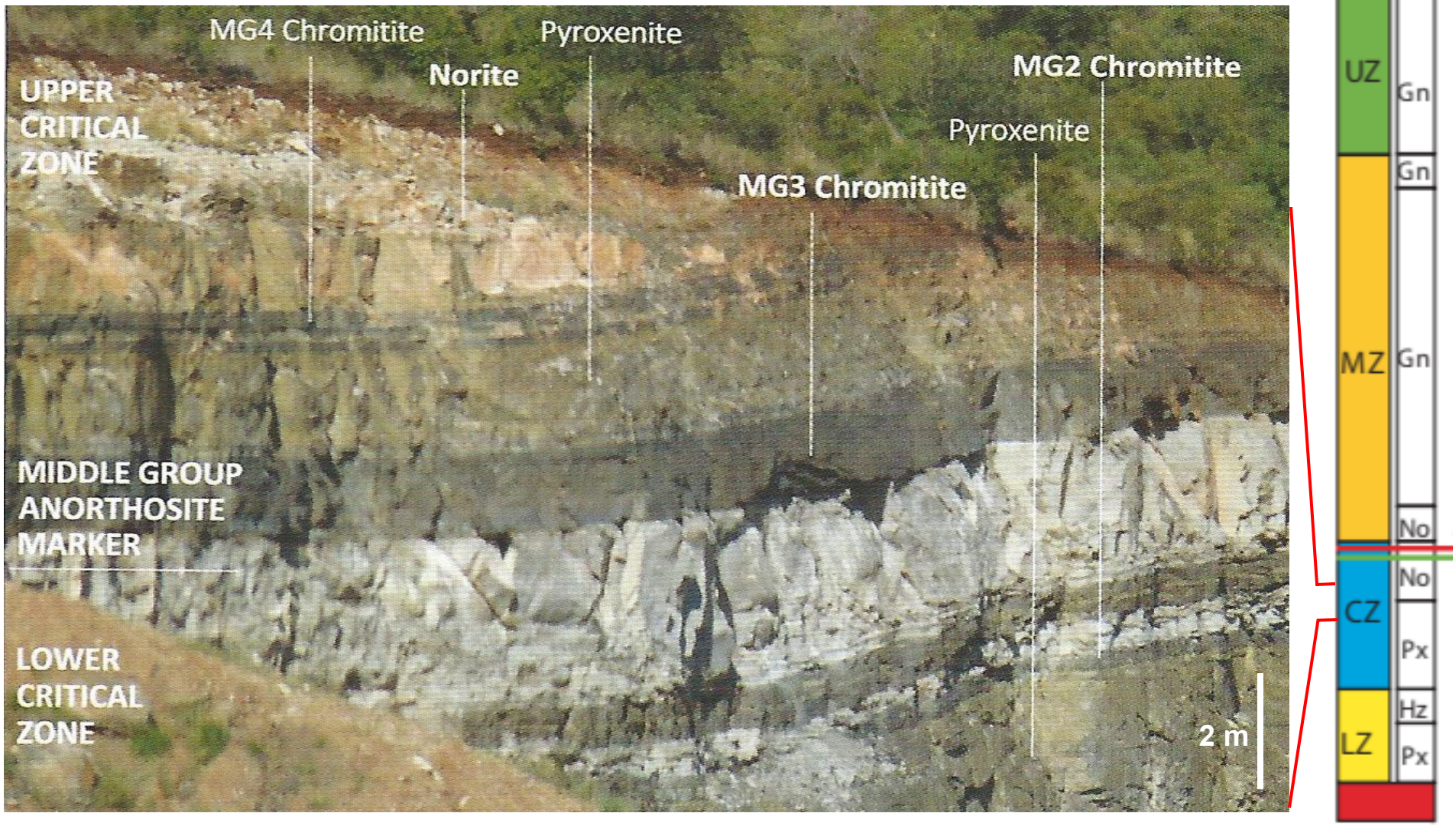


Scoon 2016



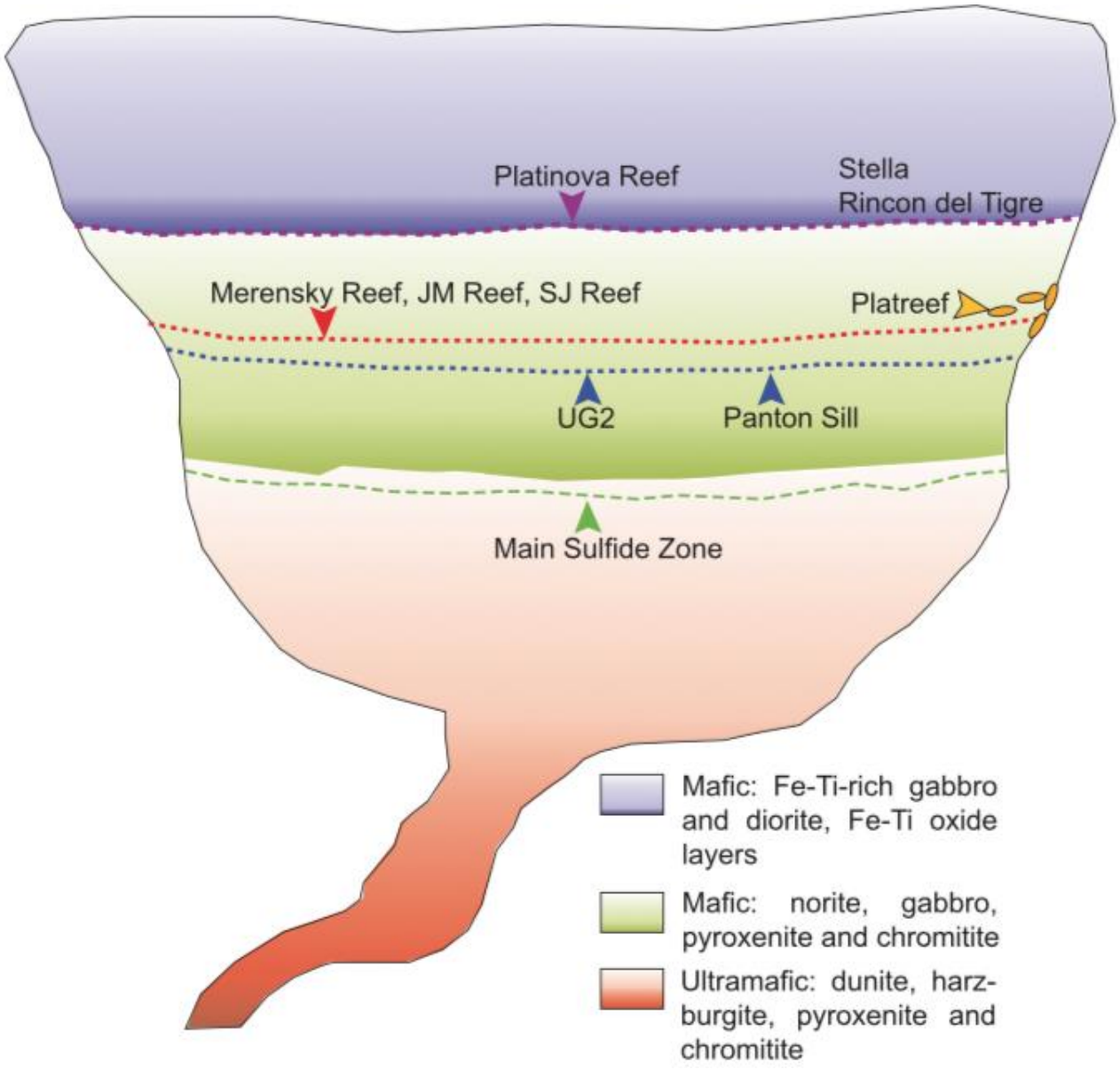
Scoon 2016

Bushveld Complex
Stratigraphy



Scoon 2016

Bushveld Complex
Stratigraphy

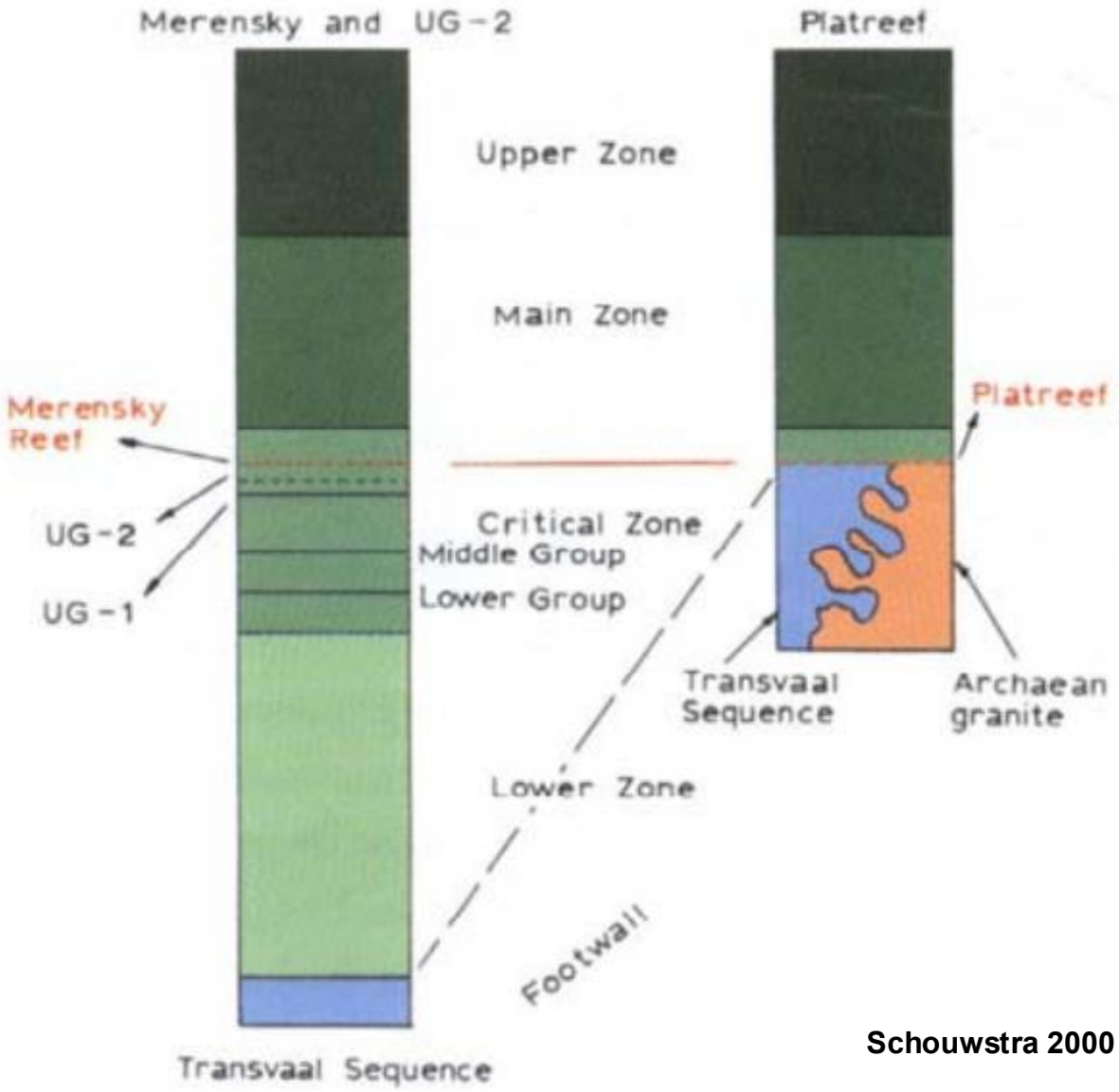


Gabbroic cumulates: First feldspar

Ultramafic cumulates

Godel 2015

PGE ores
Ore body stratigraphy

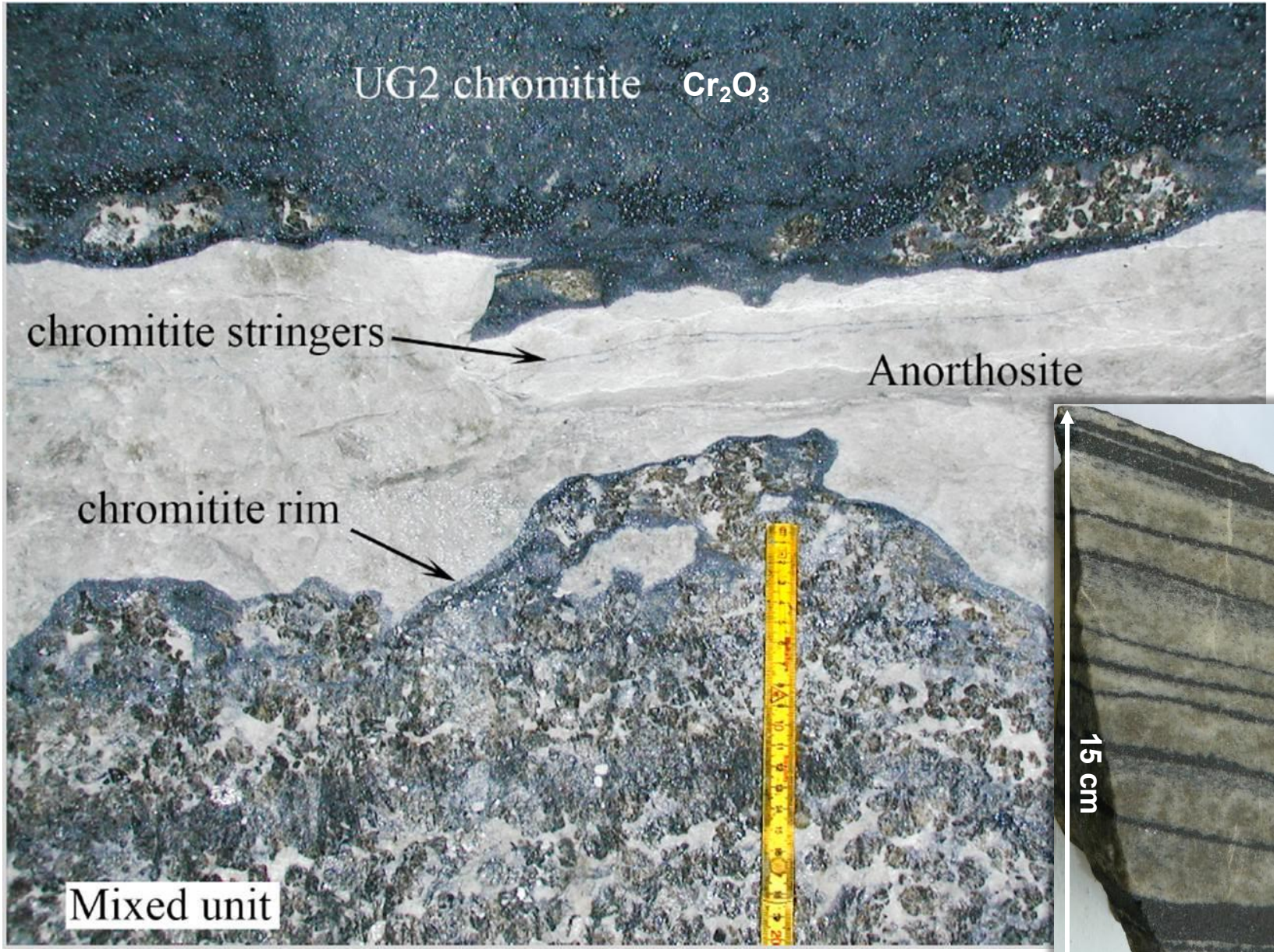


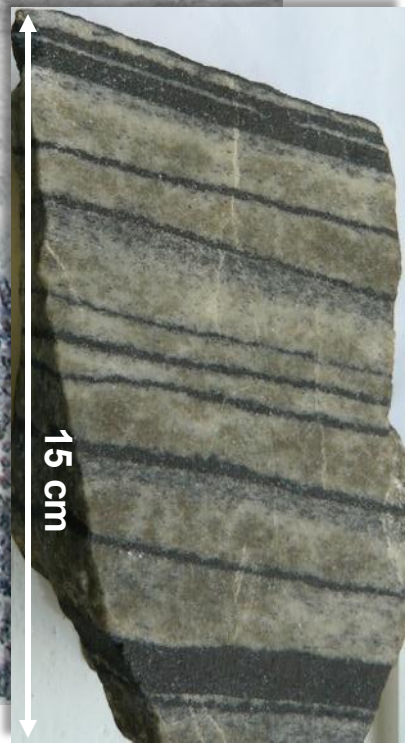
Schouwstra 2000

- **Main PGE sources:**
  - **Merensky Reef (since 1926)**
  - **UG-2 (main today)**
  - **Platreef**
- **Reefs are 0.4-2.5m thick**
- **Mineralogy: pyroxene, feldspar, chromite, sulfides, PGM (calc-silicates in Platreef)**
- **Grades:**
  - **Merensky: 3-20 g/t PGE**
  - **UG-2: 4-7 g/t PGE**

PGE ores
Mineralogy

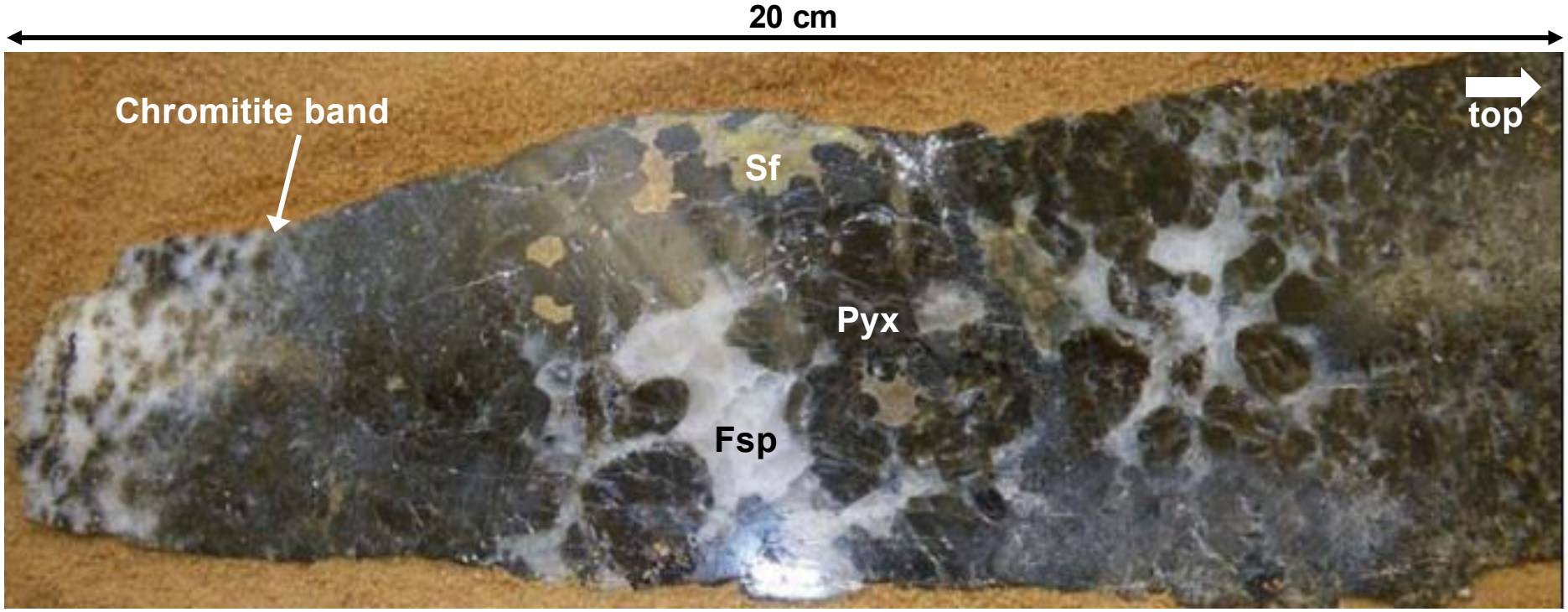
Mathez (2005)





PGE ores
Mineralogy

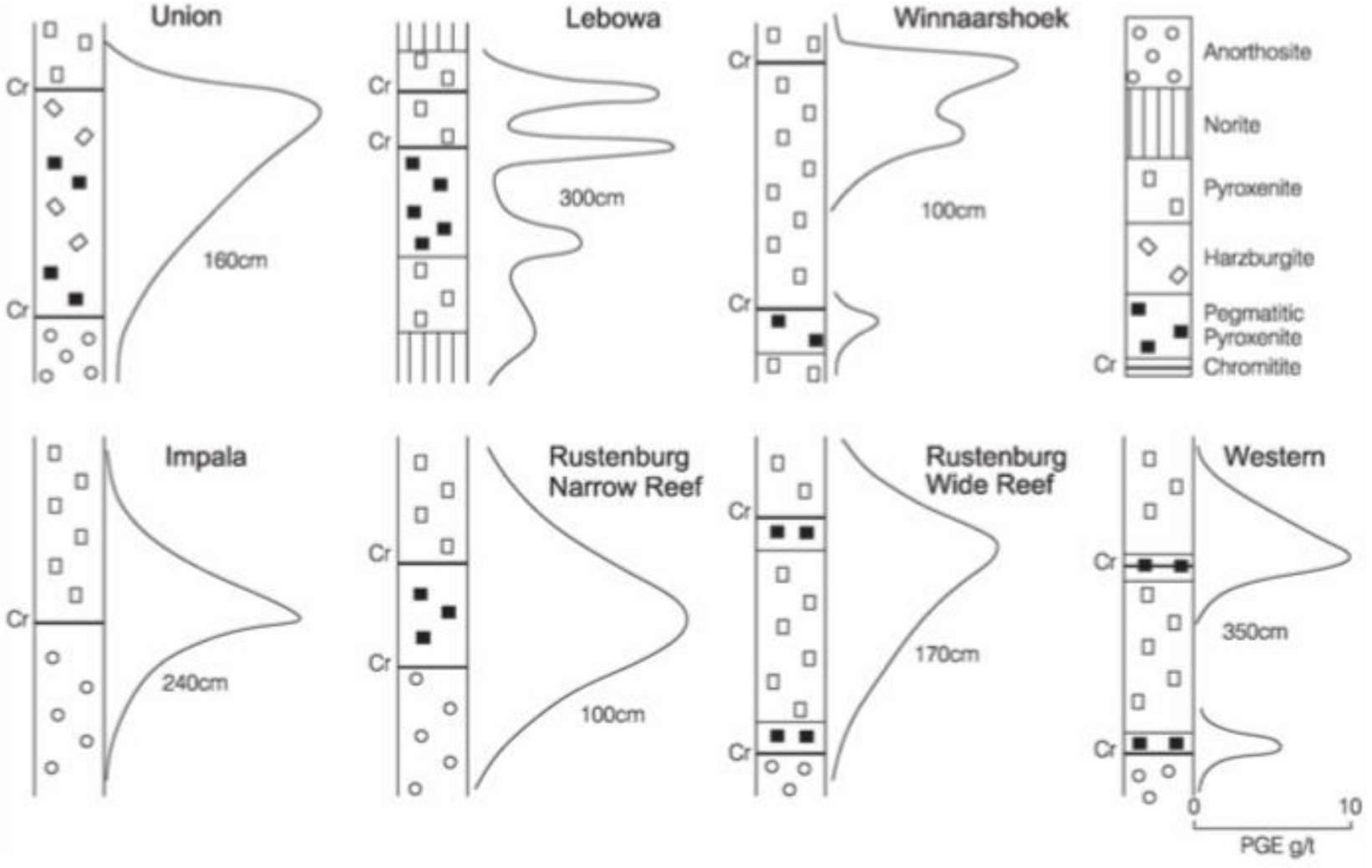
Merensky Reef



Anorthosite **Pegmatitic pyroxenite** **Cawthorn 2010**

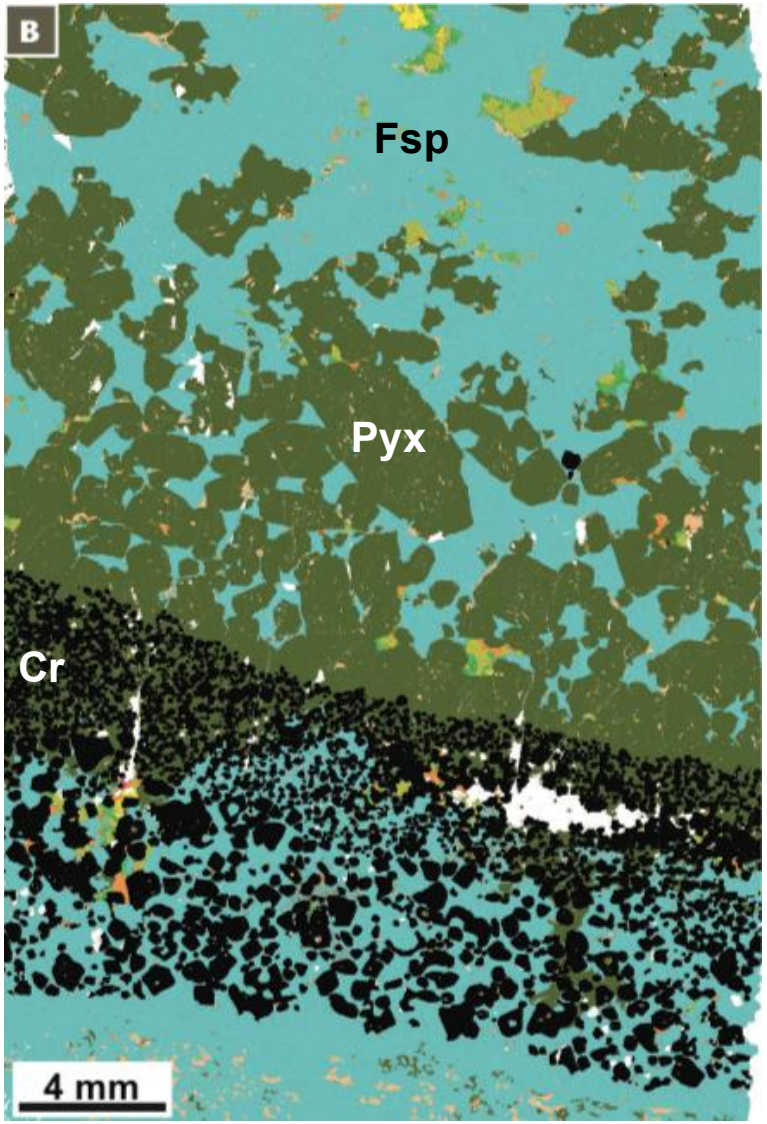
PGE ores
Mineralogy

Merensky Reef sections



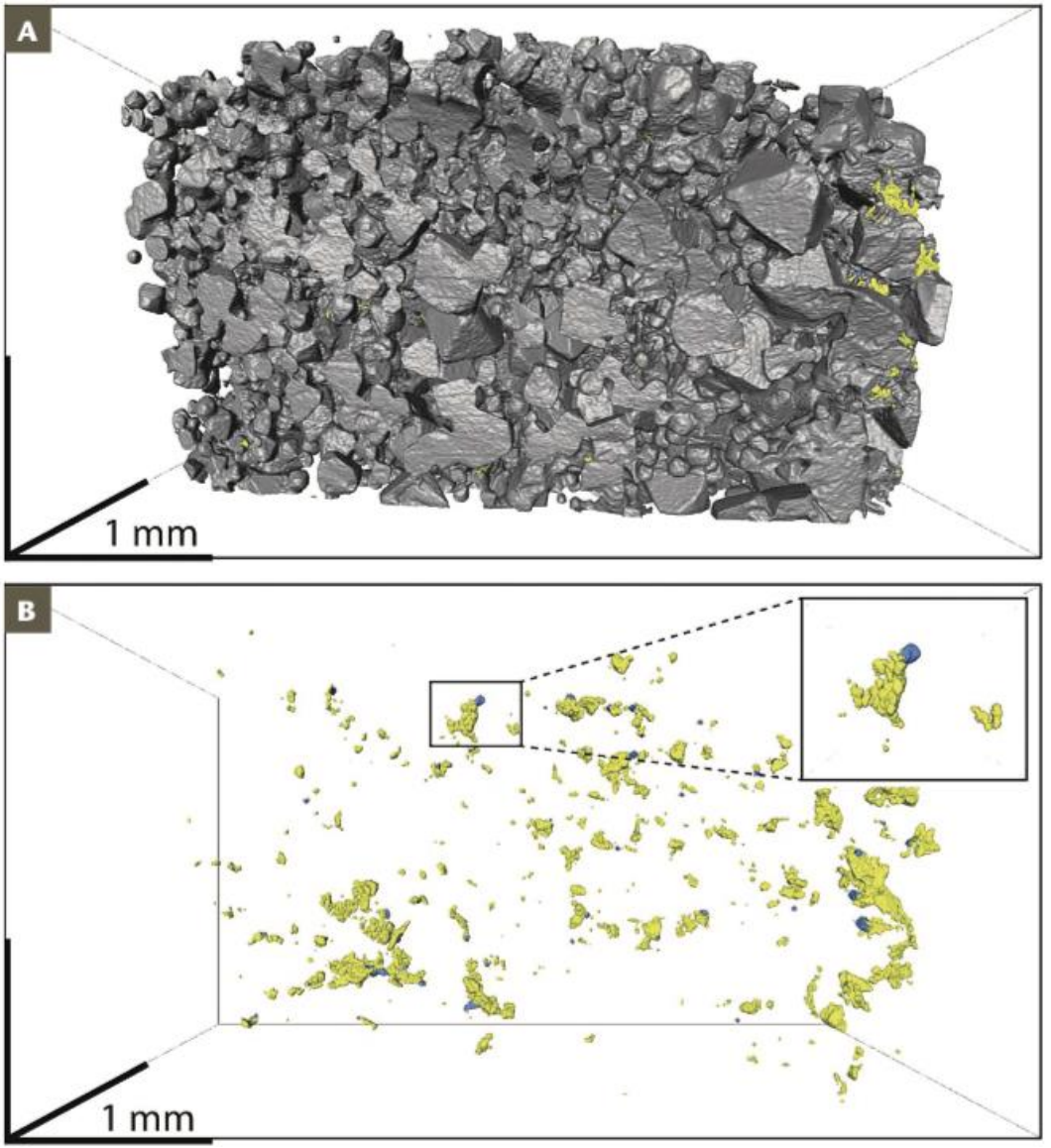
Cawthorn 2015

PGE ores
Mineralogy



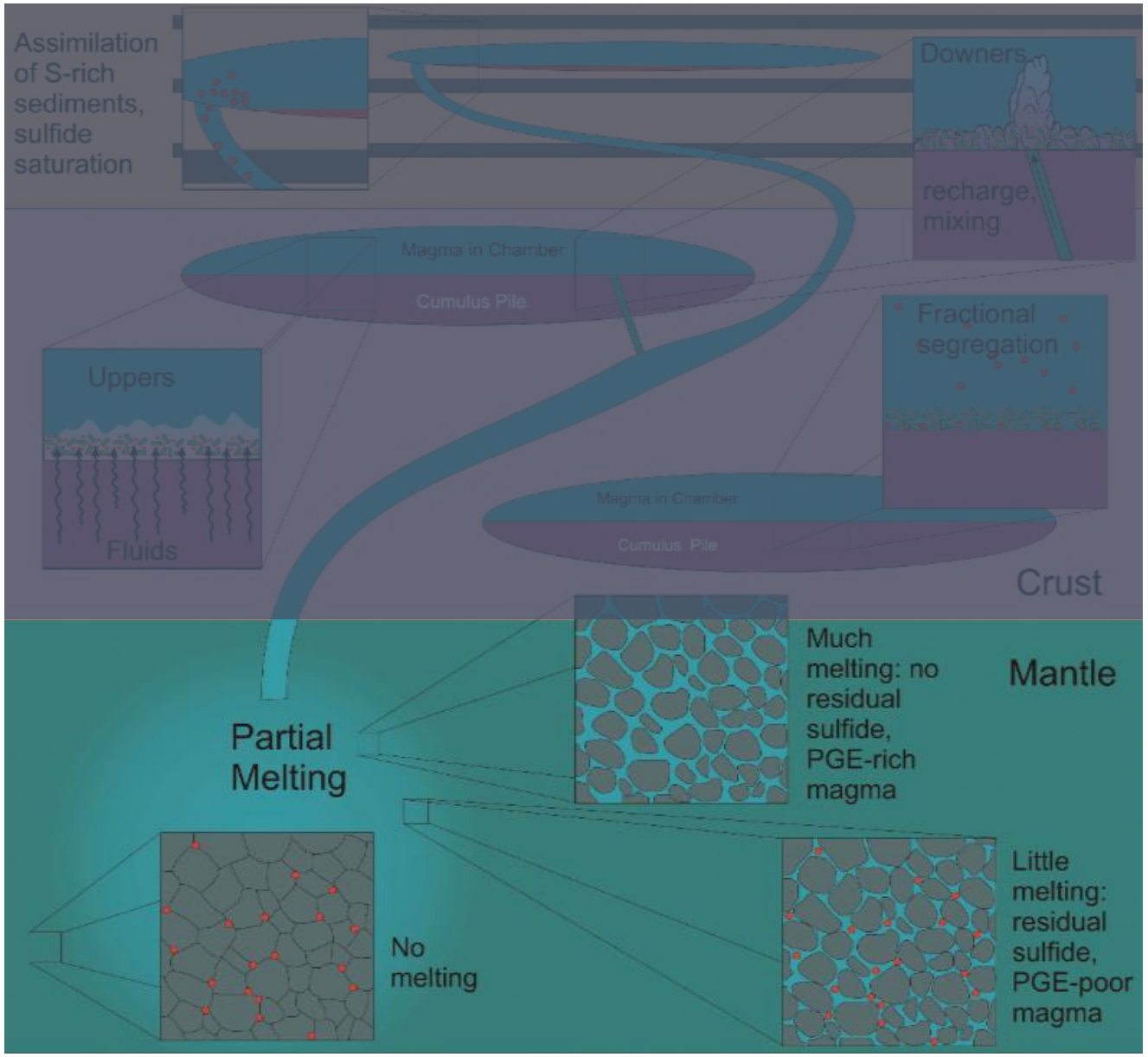
O'Driscoll 2017

Merensky Reef PGE



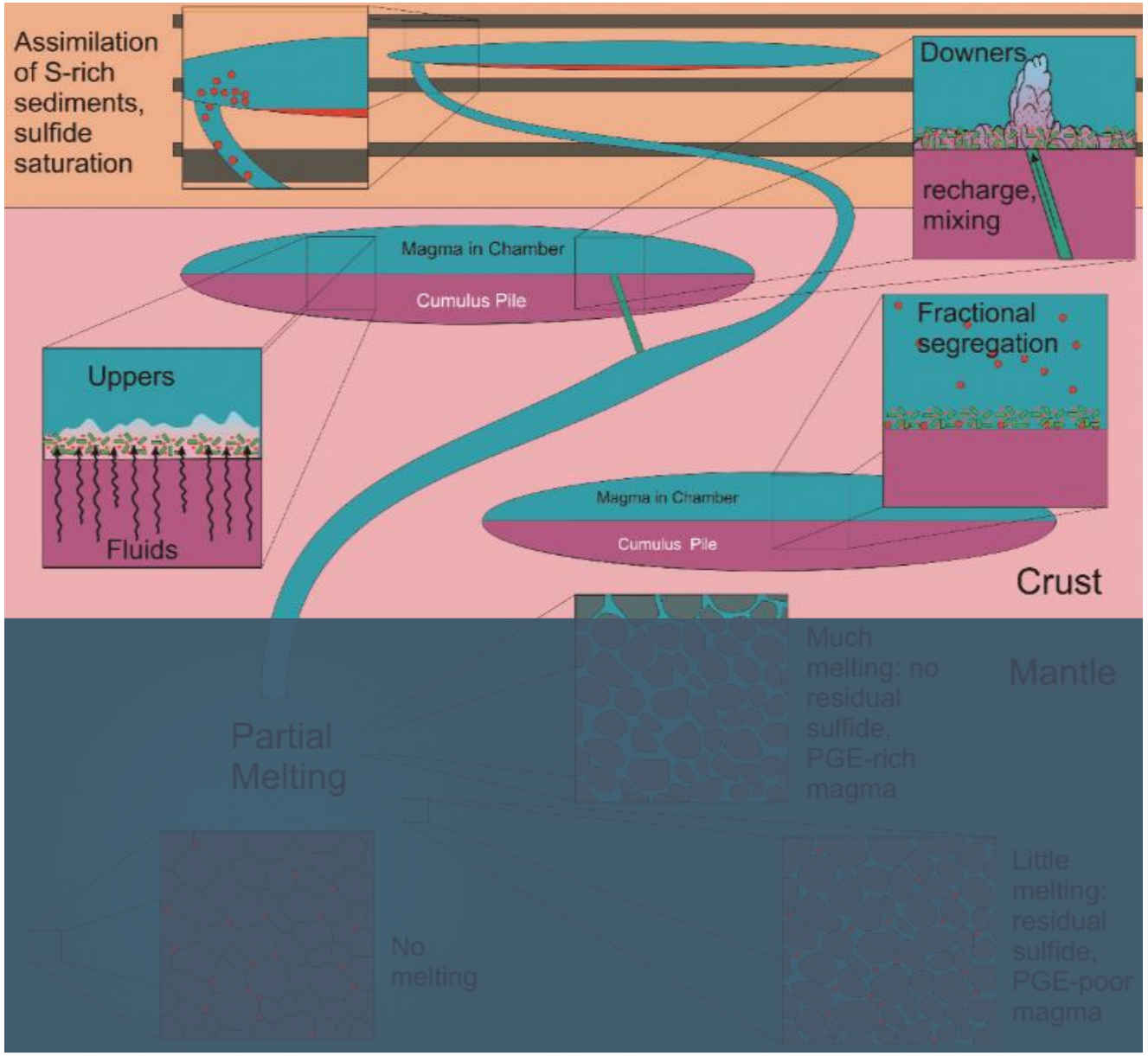
© B. Godel

PGE ore genesis and magmatic processes
Partial melting



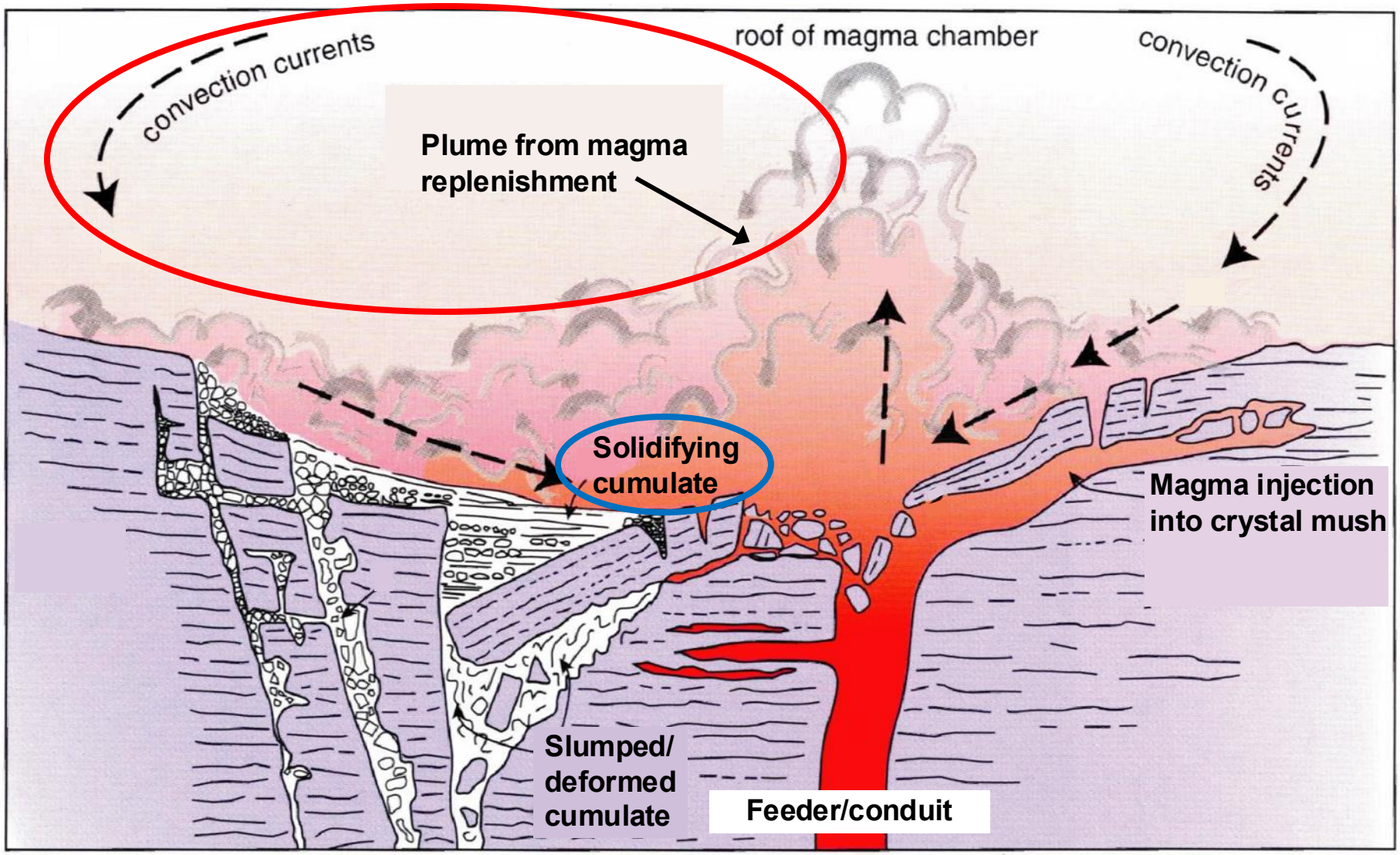
Mungall 2008

PGE ore genesis and magmatic processes
Partial melting



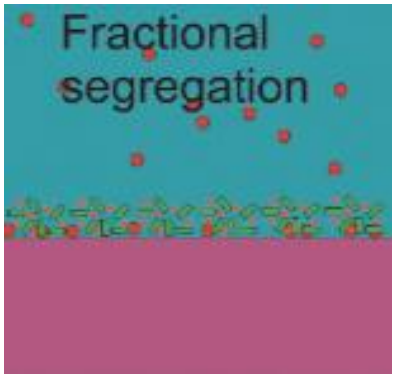
Mungall 2008

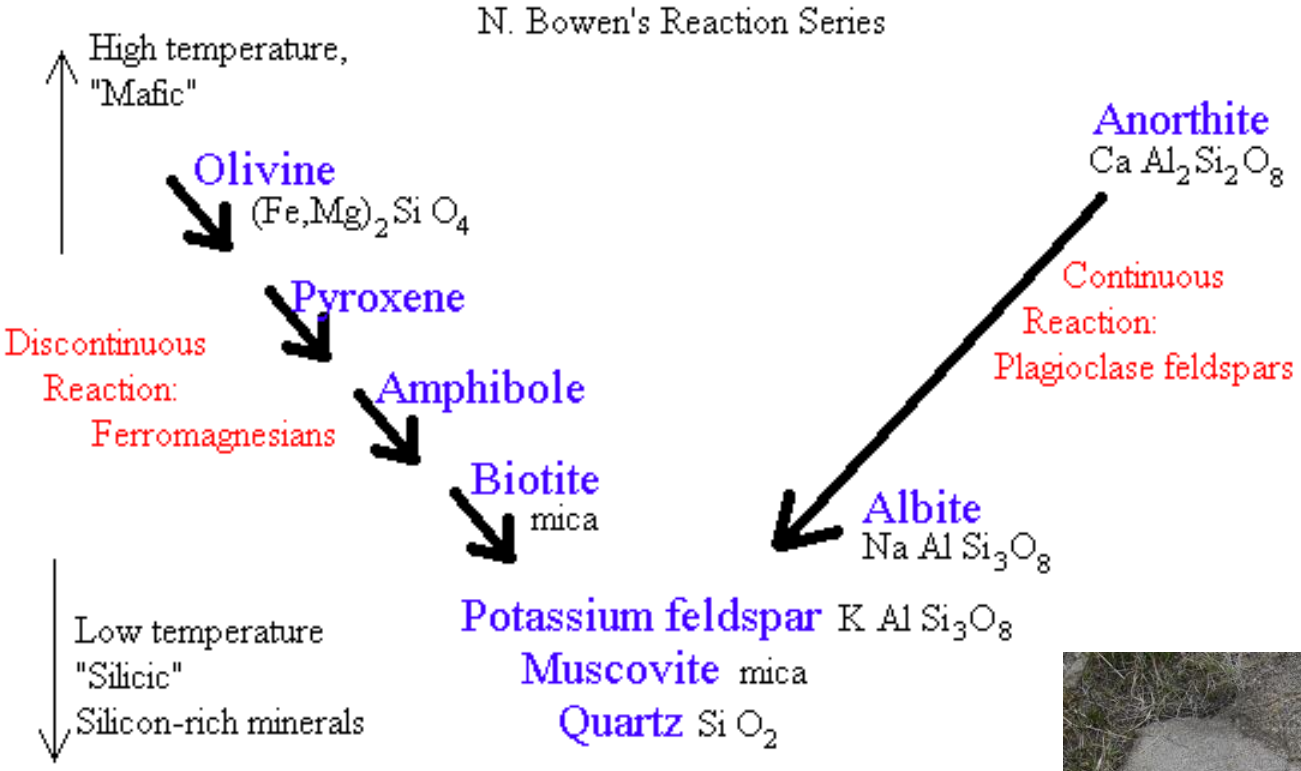
PGE ore genesis and magmatic processes
Magma chamber processes



Emeleus 1996

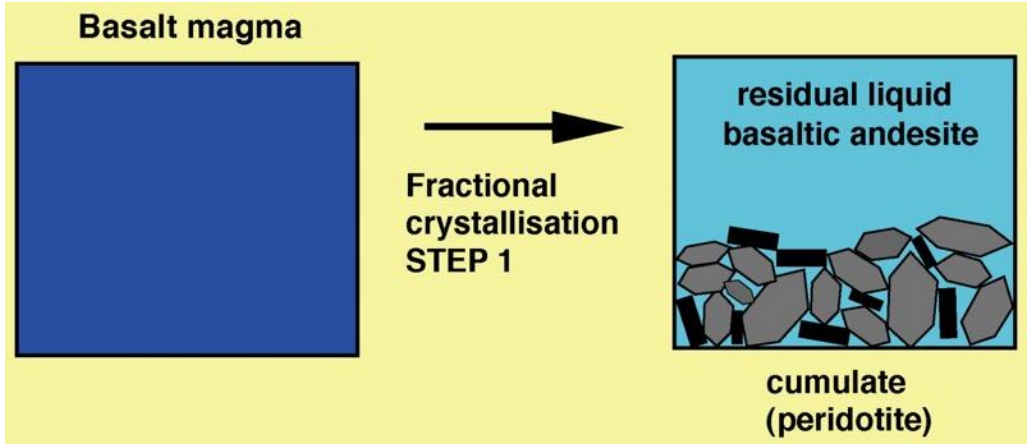
PGE ore genesis and magmatic processes
Crystallization + cumulates





Stoke's Law:

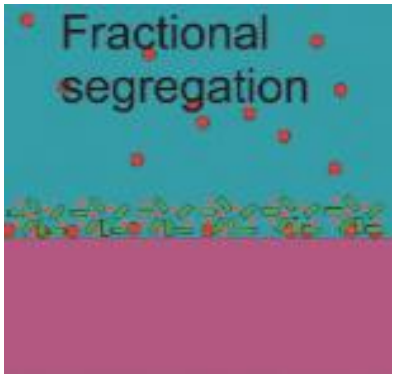
$$V = \frac{(\rho_1 - \rho) \times g \times d^2}{18 \times \mu}$$

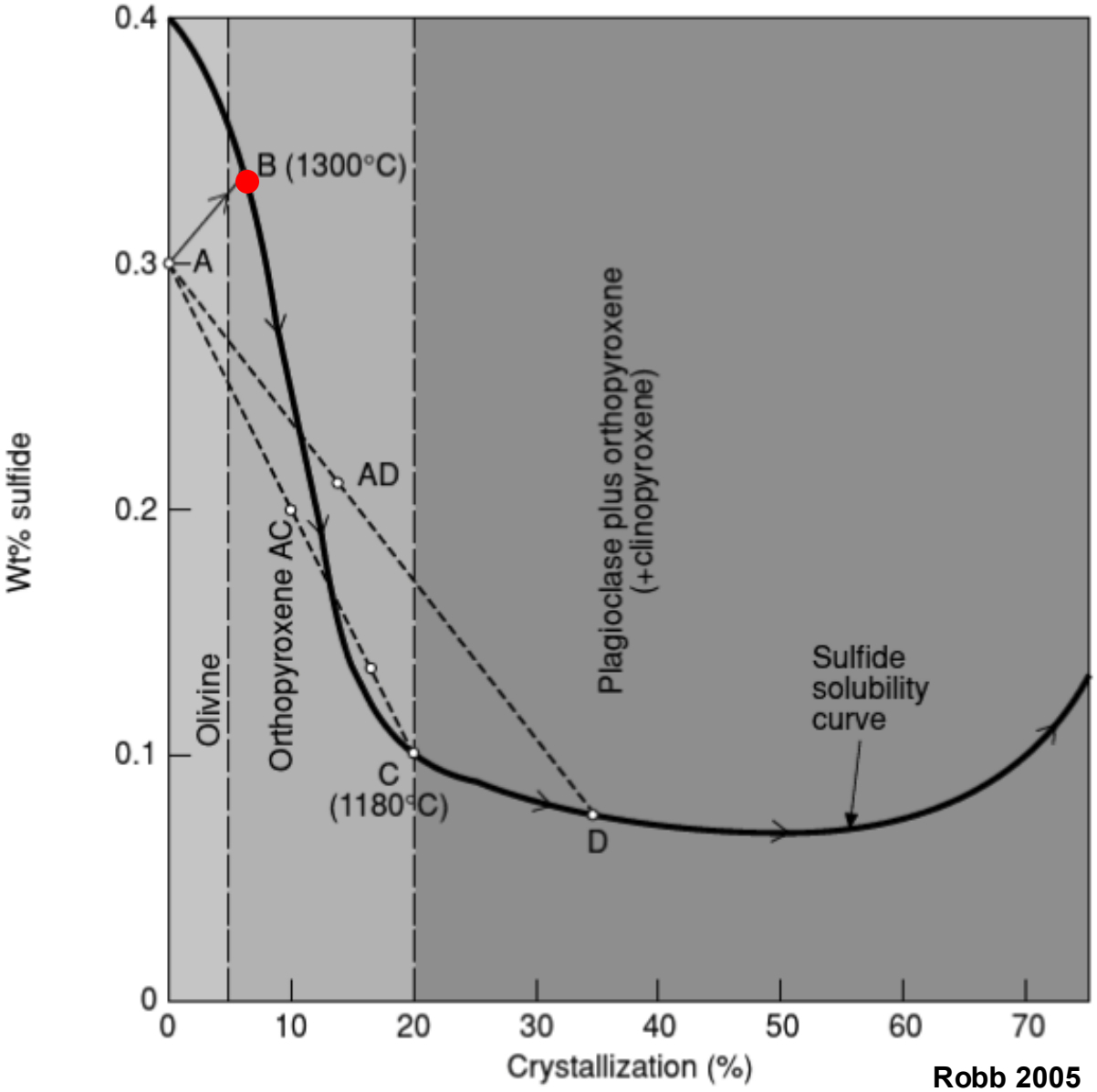




© B O'Driscoll

PGE ore genesis and magmatic processes
Crystallization processes, sulfide saturation

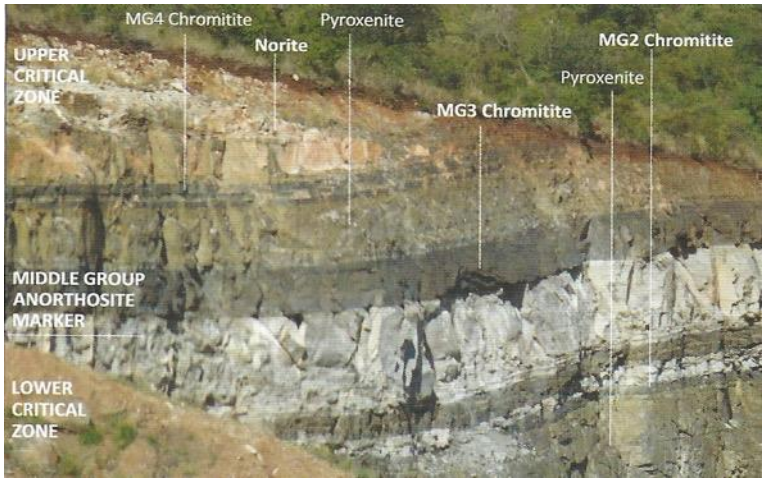




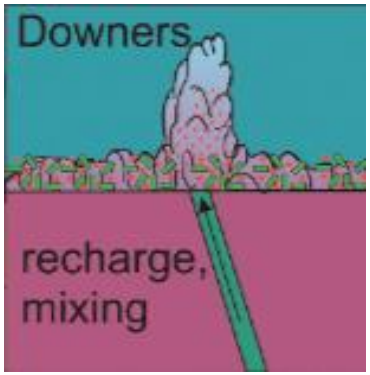
Robb 2005

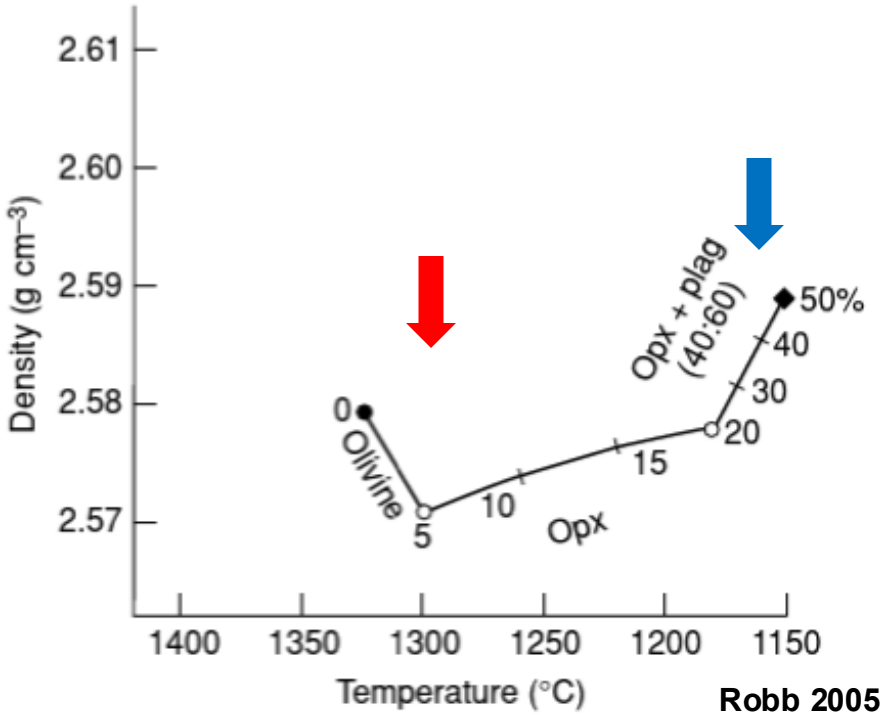
- Fractional crystallization -> cumulate formation and sulfide saturation
- PGE adhere to sulfide droplets and clusters

PGE ore genesis and magmatic processes
Magma mixing

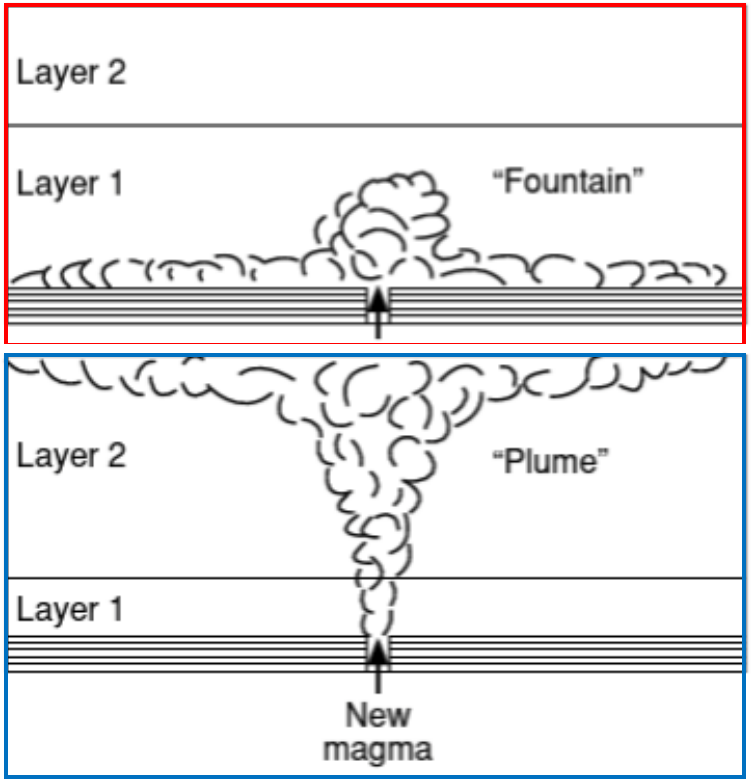


- Magma injection forms layering
- Magma injection promotes:
  - magma mixing and PGE scavenging by sulfide droplets
  - interruption of normal crystallization, formation of mono-mineralic chromite seams

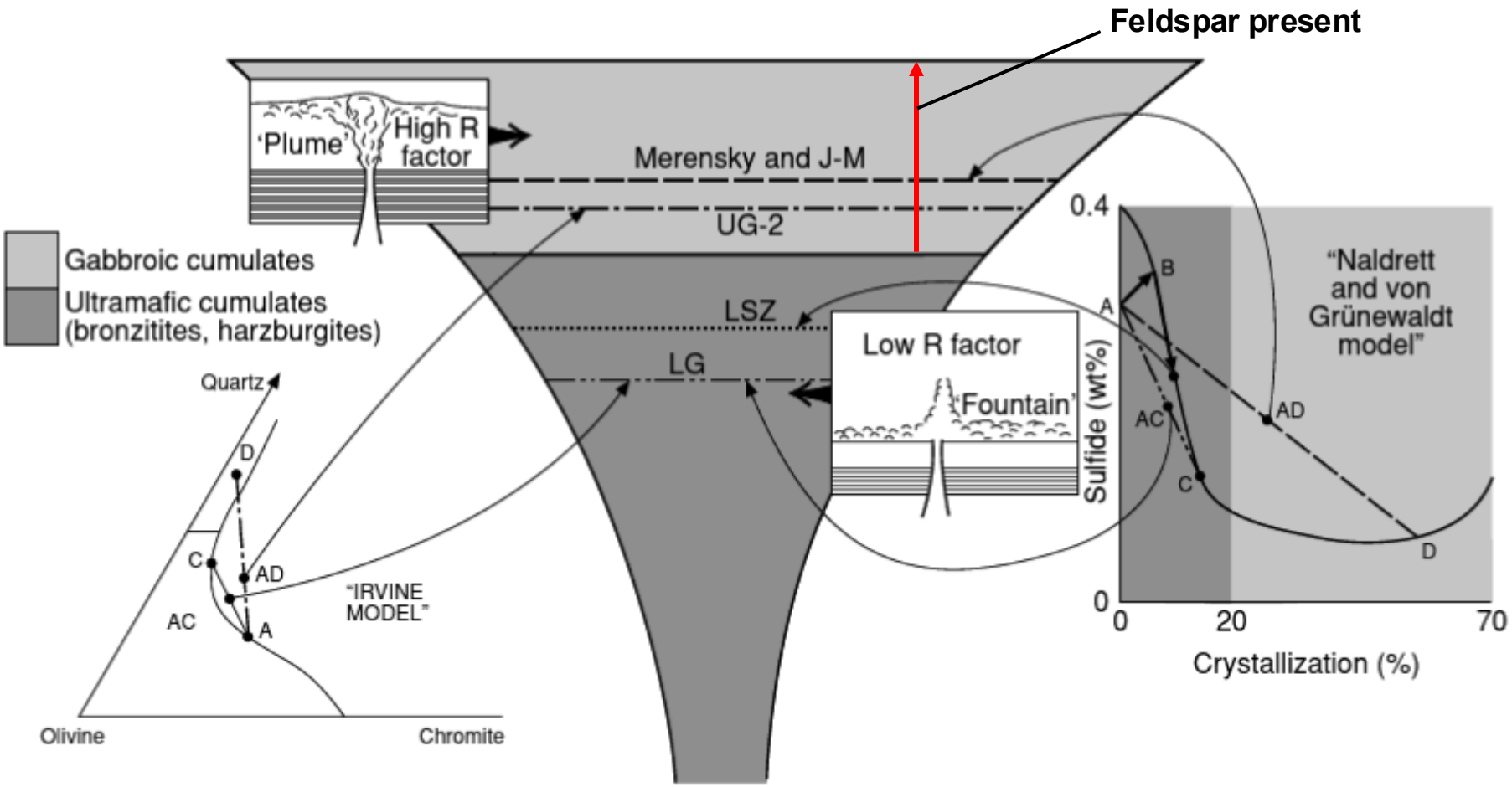




Robb 2005



PGE ore genesis and magmatic processes
Model



Robb 2005

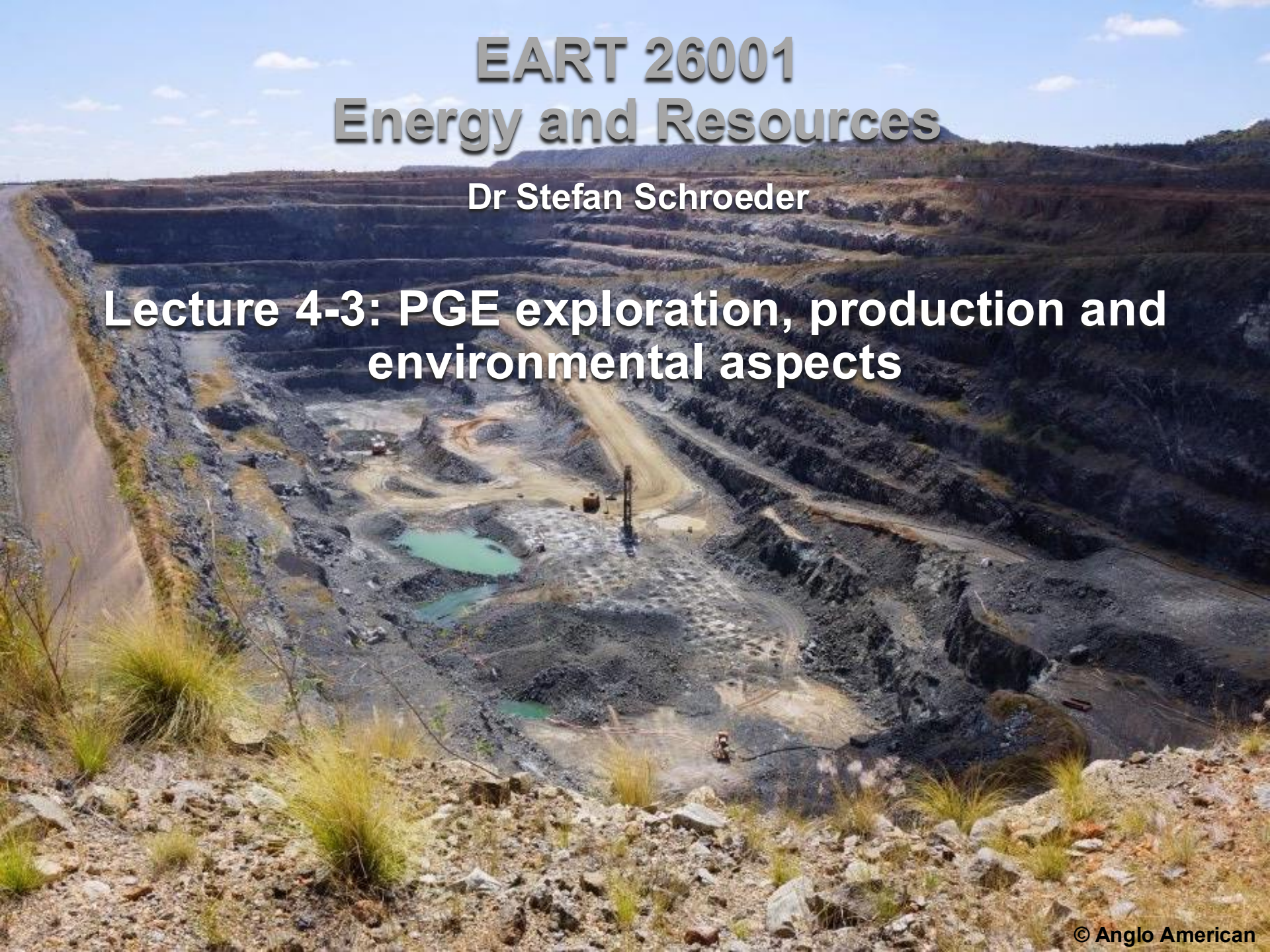
PGE ore deposits and magmatic processes
Summary

- The Bushveld Complex is the largest layered intrusion in the world, and currently the main source of PGEs.
- Magmatic processes of PGE ore genesis are strongly tied to processes of partial melting and crystallization in a magma chamber.
- PGEs bond with iron and sulfide minerals, and the formation of sulfide melt droplets to scavenge PGEs from melt is a key factor in ore genesis. This can be achieved by several mechanisms:
    - Density segregation of sulfides
    - Magma re-injection and mixing
    - Assimilation of and reaction with (S-rich) country rock
    - Later alteration by aqueous phases
- Layered intrusions are also important sources of Cr, Ni and V.

EART 26001
Energy and Resources

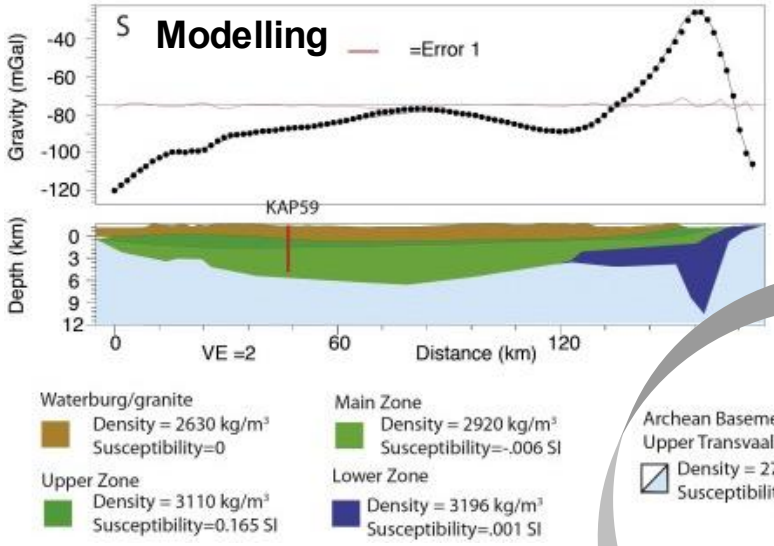
Dr Stefan Schroeder

Lecture 4-3: PGE exploration, production and environmental aspects

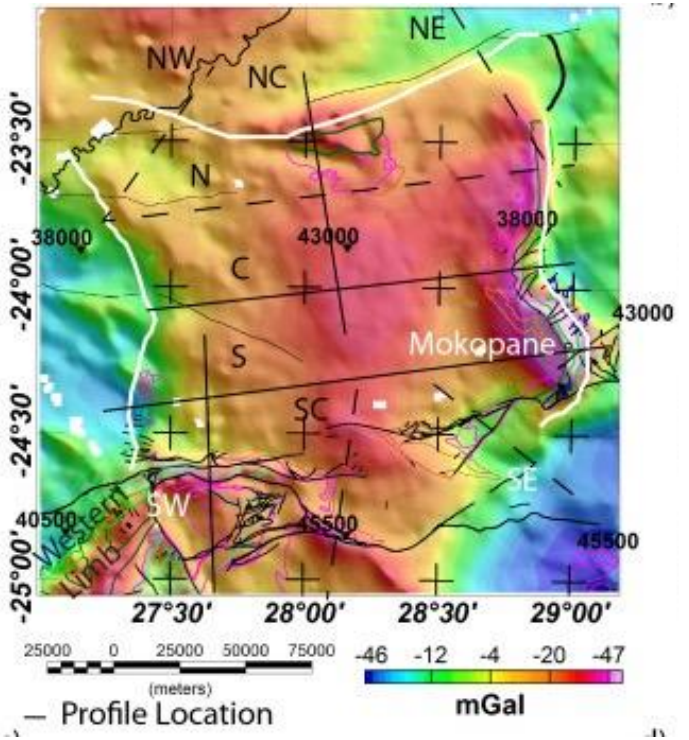


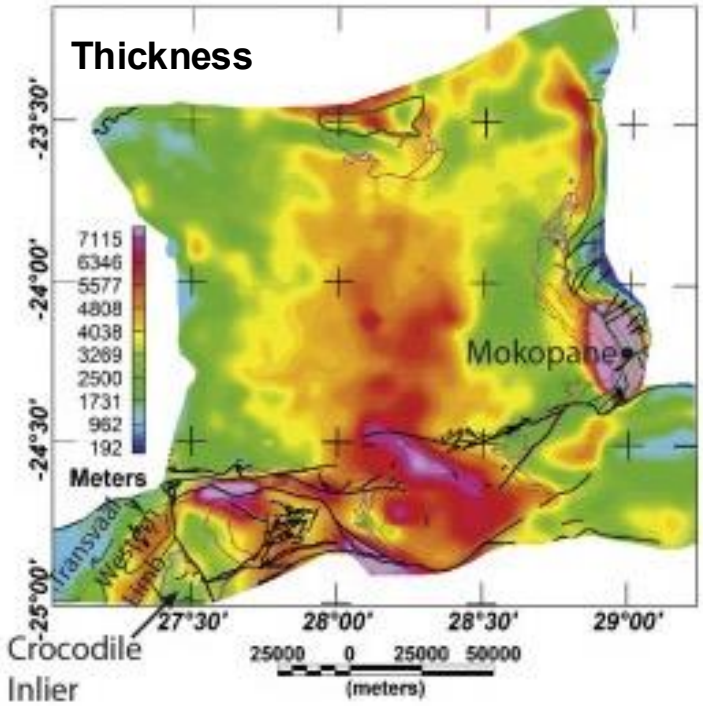
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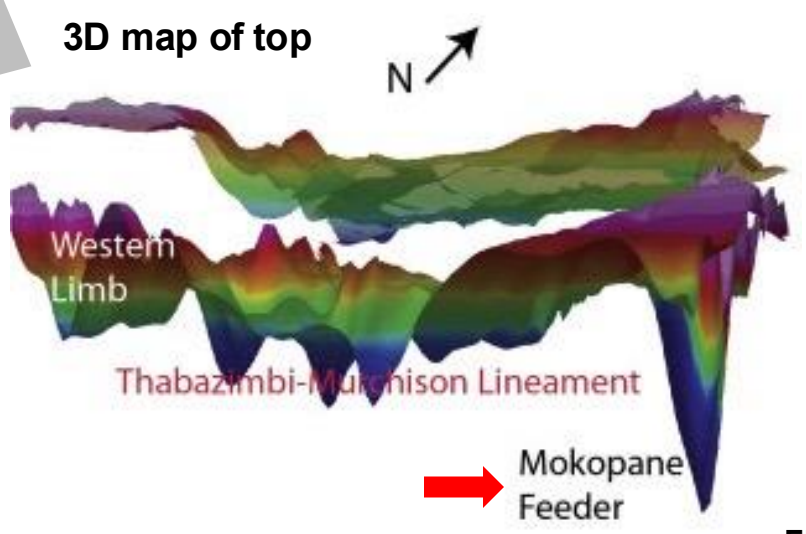
Exploration
Geophysical techniques – gravimetry



Density anomalies



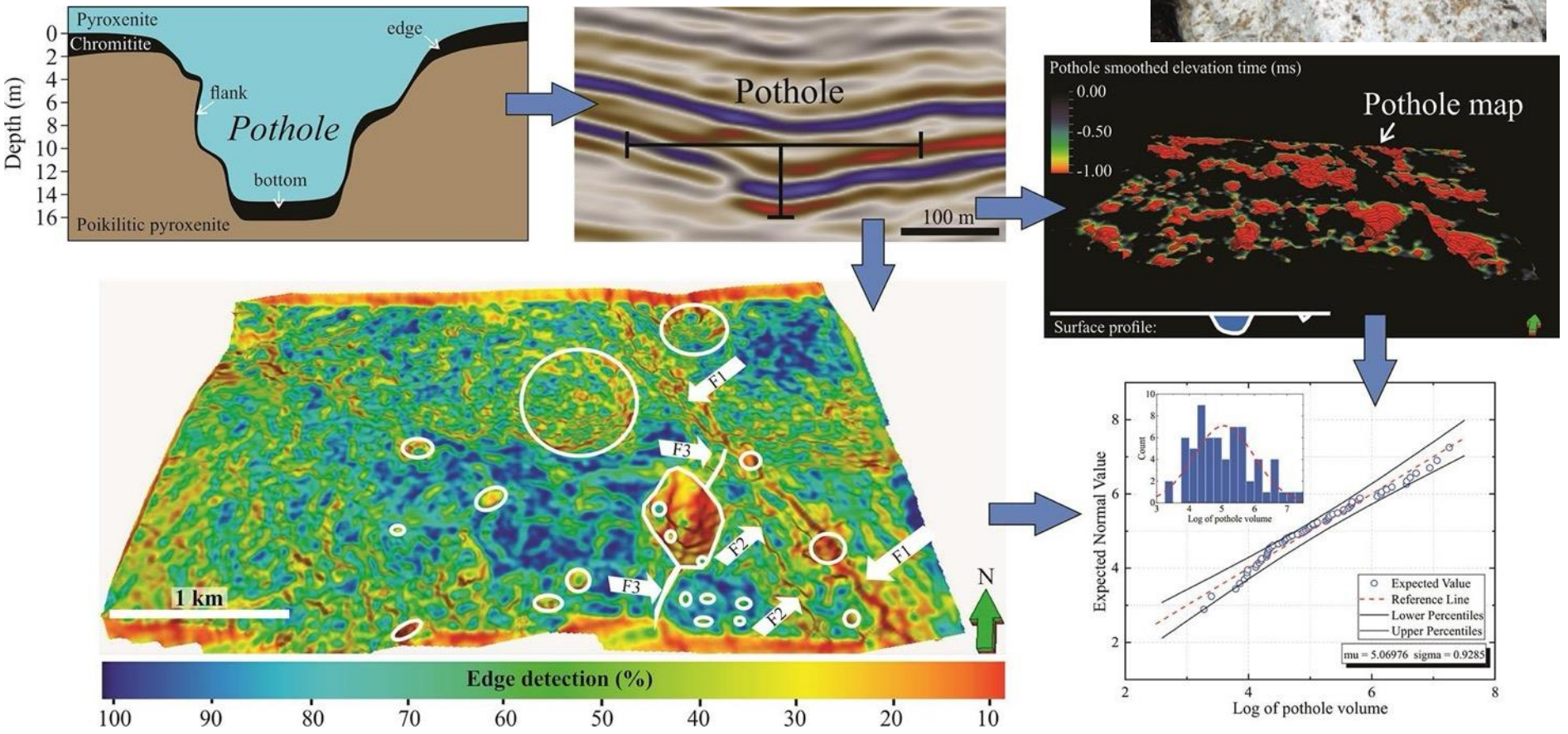




Finn 2015

Exploration
Geophysical techniques - seismic

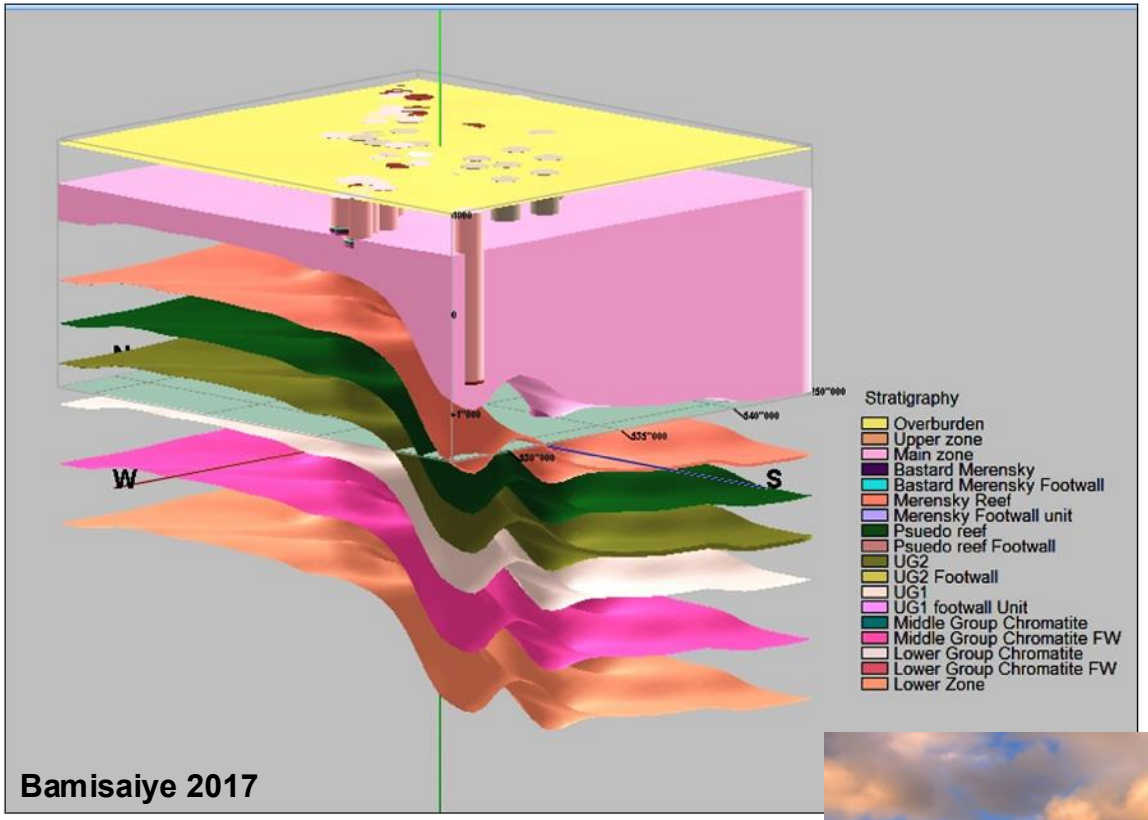




Sehoole 2020

- Seismic survey allows recognition of features >25m in size
- ‘Potholes’ constitute significant ore loss (~15%)

Production
Mining and Beneficiation



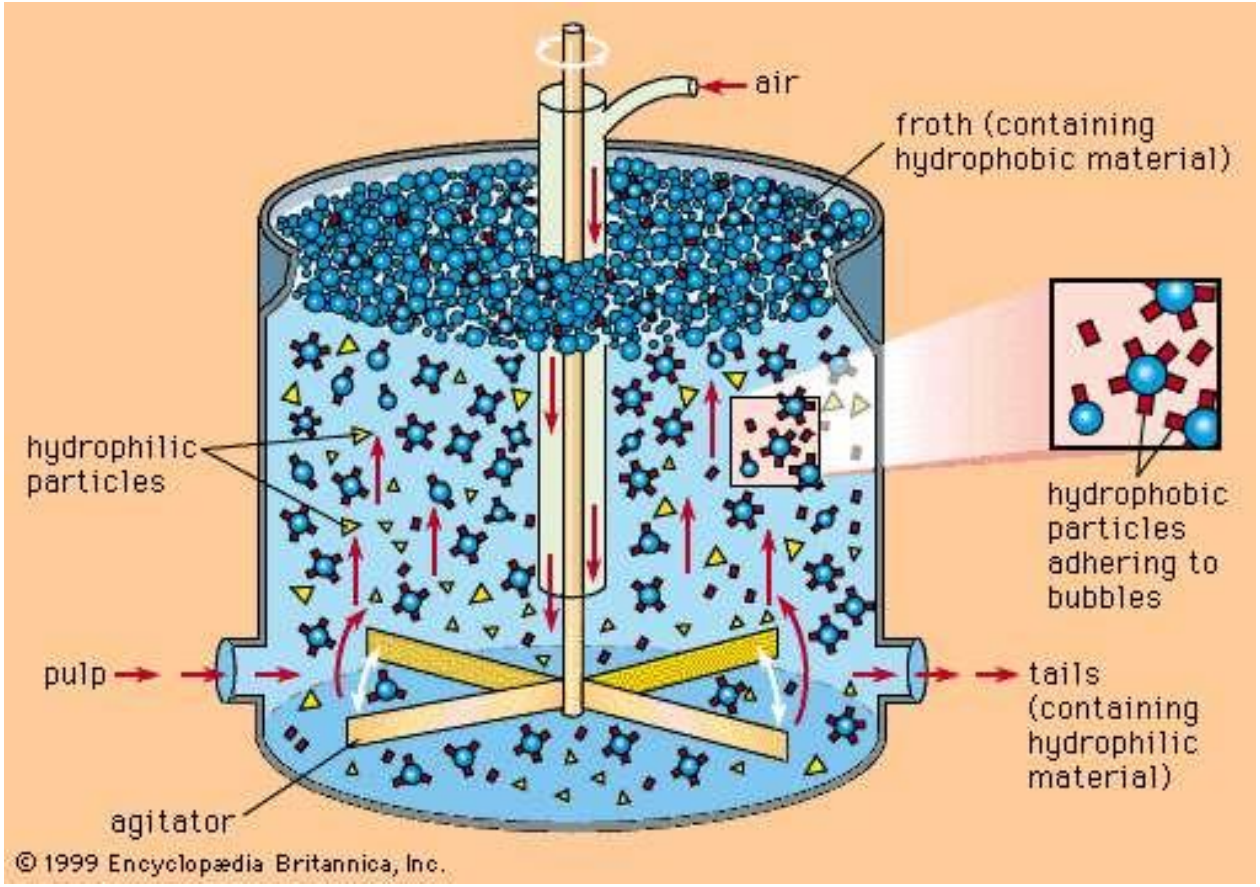
Bamisaiye 2017

- Underground mining 500-2000m, increase of open pit
- Conditions in deepest mines similar to Witwatersrand
- Processing via flotation and electric smelters
- Concentrate 0.02-0.2% PGE = 5 x 10^5-10^6 crustal values


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Production
Mining and Beneficiation

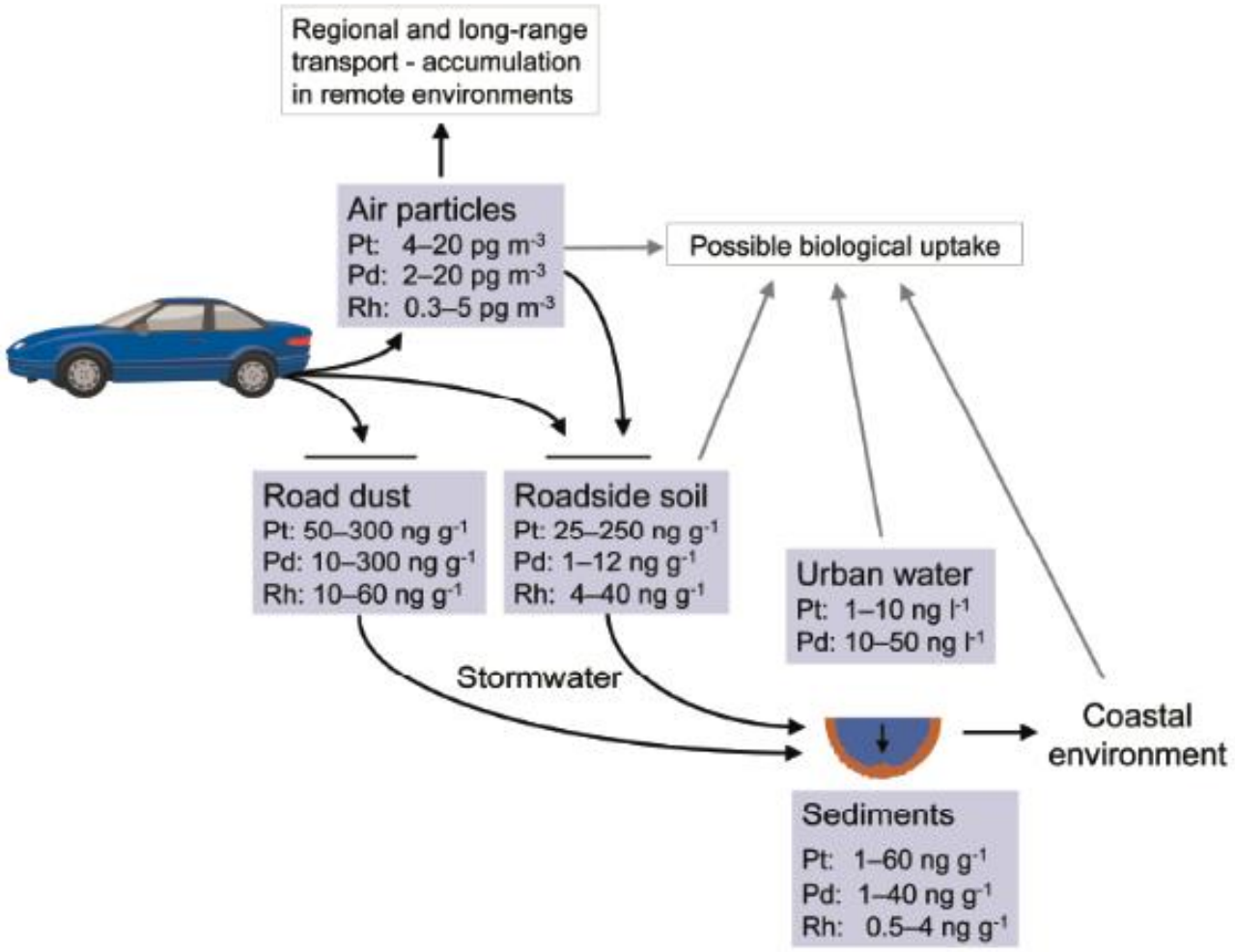
Flotation



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Environmental aspects
PGE in environment

- **Main emission sources: cars, smelters, medical facilities**
- **Nanoparticles**
- **Largely inert and low concentrations, but increasing evidence for bioavailability (toxic effects?)**



Rauch 2008

Platinum group element deposits
Summary week 4

- Platinum group elements (PGE) are very rare in the crust and require high enrichment to ore grade.
- They are precious metals, resistant to corrosion and heat, and are widely used in catalytic applications, electronics/electrical applications, medical applications and as jewelry. They are considered critical resources.
- The main deposit type are layered igneous intrusions where orthomagmatic processes concentrate PGEs.
- Bonding with iron and sulfide phases and segregation of these from the parent melt allows local strong enrichment of PGEs to ore grade.
- Potential environmental impact of PGEs is little studied.